MDANSE for neutron scattering analysis

Maciej Bartkowiak, Chi Y. Cheng, Sanghamitra Mukhopadhyay

MDANSE Workshop 2026

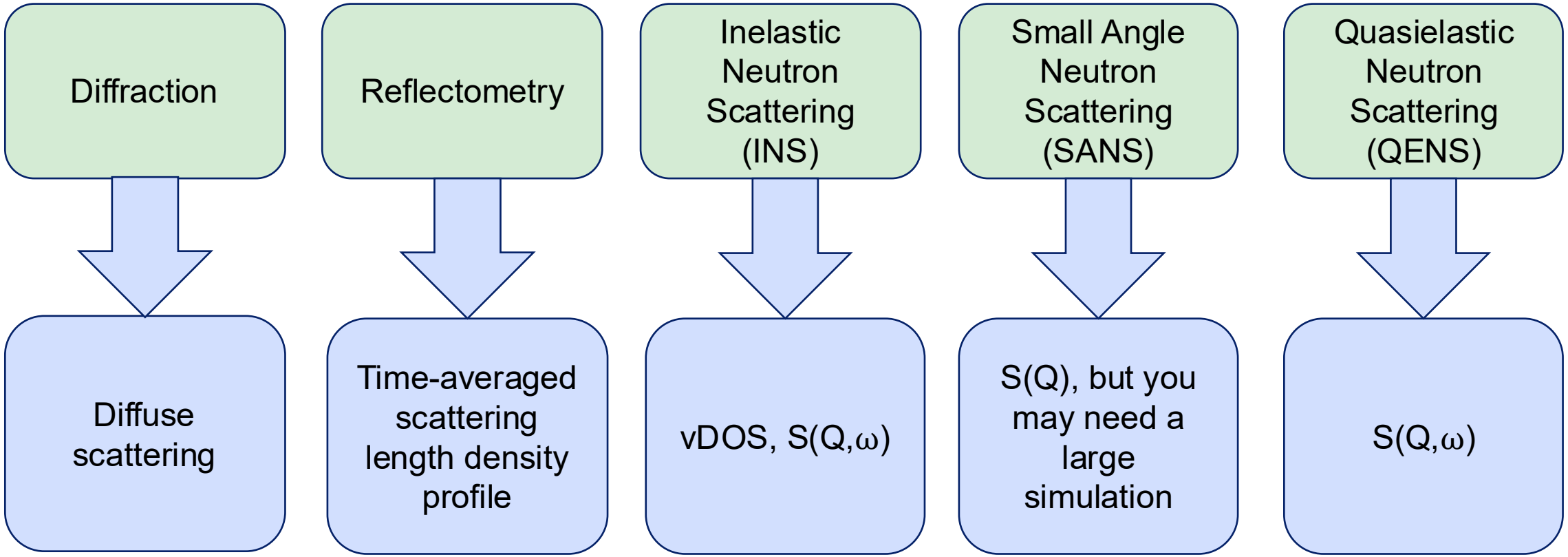
03/11/2026

Science and Technology Facilities Council
ISIS Neutron and Muon Source

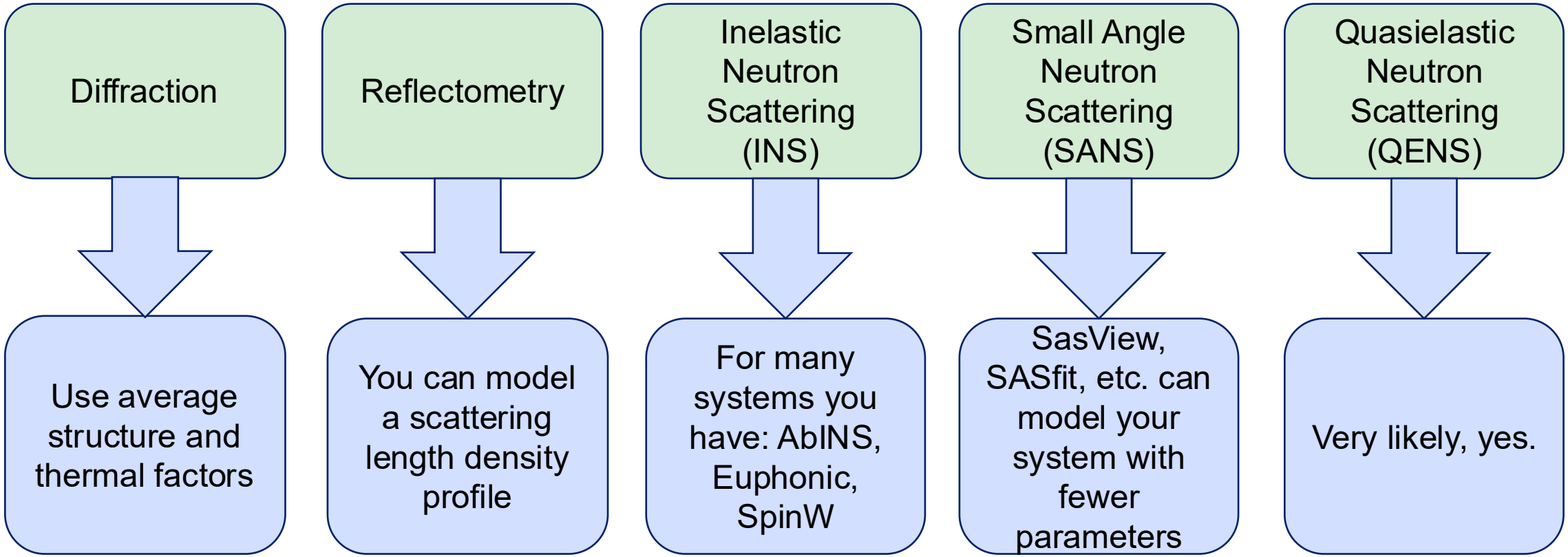
Presentation plan

1. Applications of molecular dynamics to neutron scattering techniques.
2. Simulation time and experiment energy.
3. Dynamic structure factor (coherent and incoherent).
4. Q-vector sampling.
5. Examples of neutron instrument resolution.
6. Neutron scattering length in static structure factor calculations.

Which neutron scattering techniques can benefit from MD?



Do you really need MD?



Molecular Dynamics: the boundary conditions

There are limits of what you can get from a specific MD trajectory.

Time limits:

- Step size
- Total number of steps

Space limits:

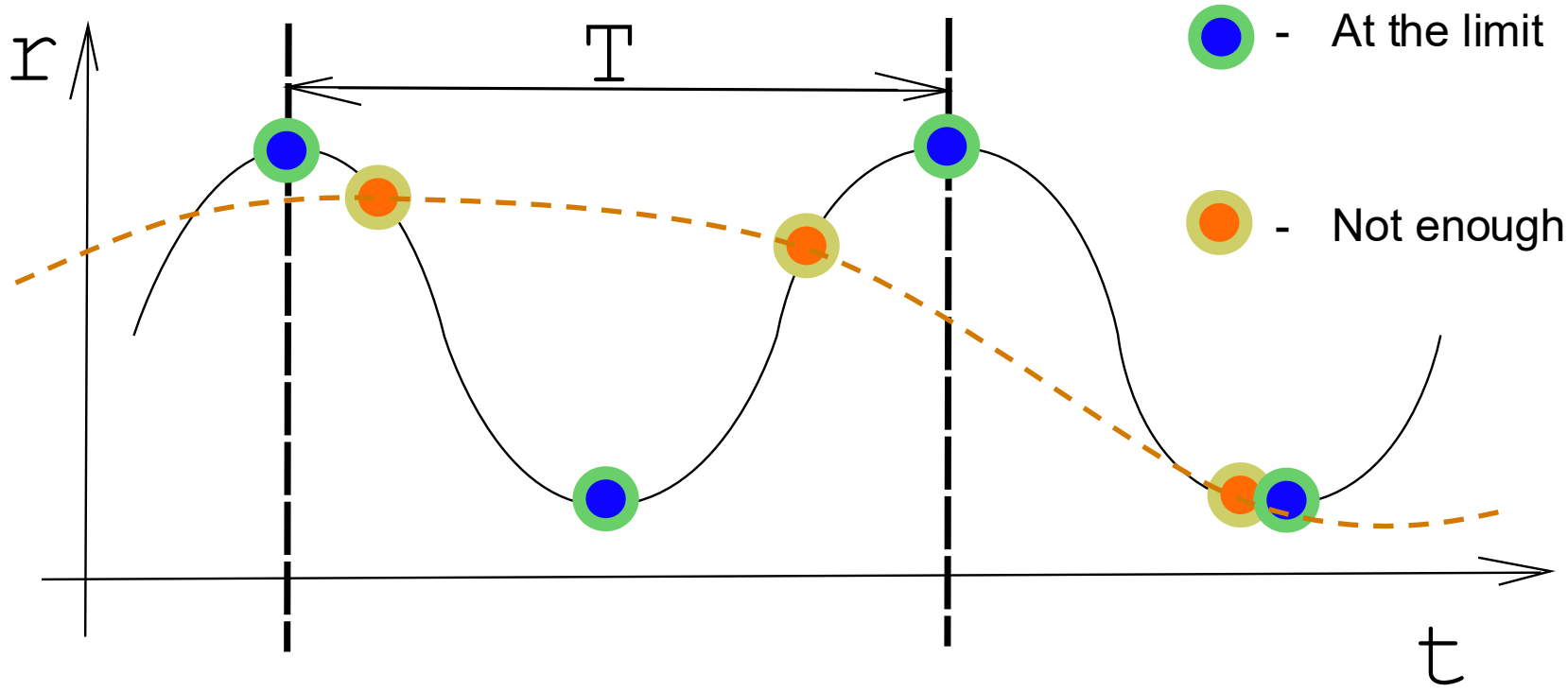
- Size of the simulation box (assuming there is one)

Molecular Dynamics: the boundary conditions

Time step vs. vibrational frequencies

Time limits:

- Step size

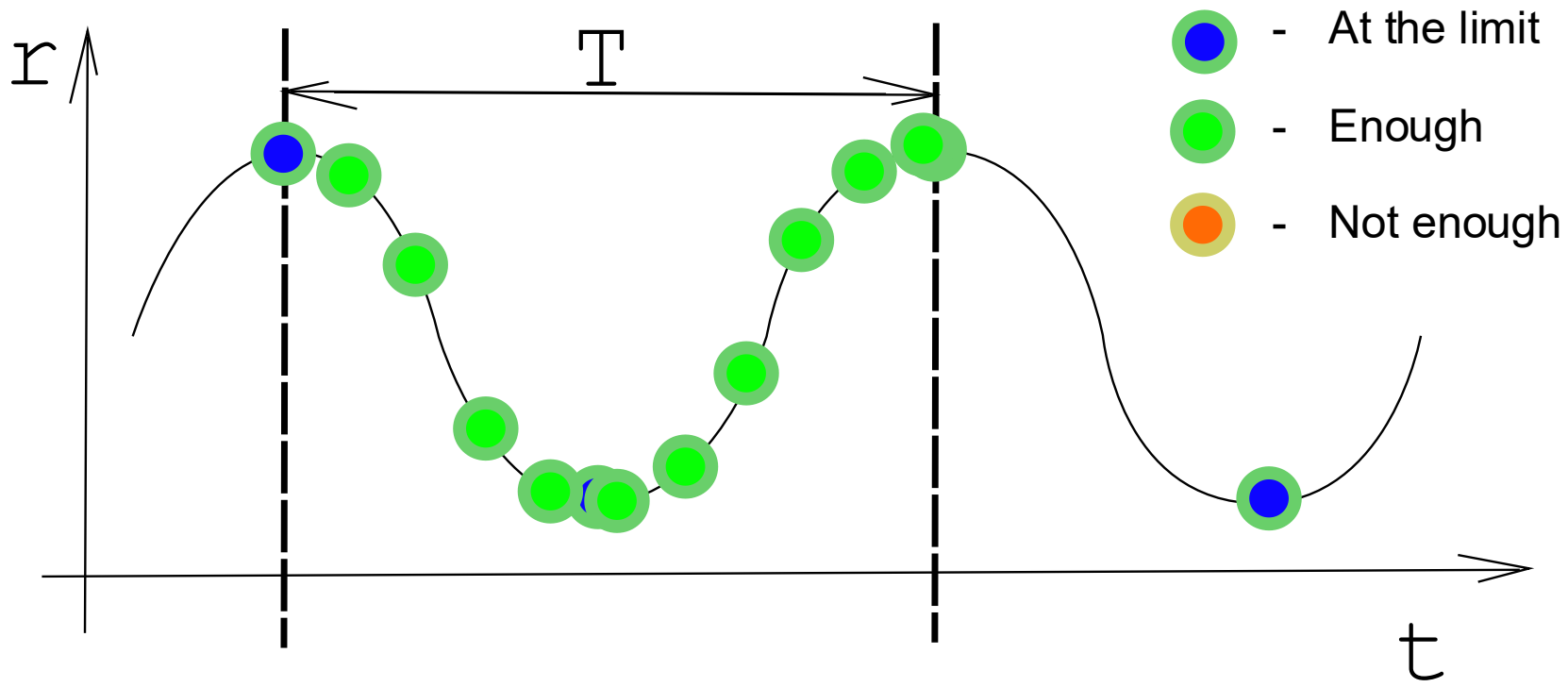


Molecular Dynamics: the boundary conditions

Time step defines the maximum energy you can access.

Time limits:

- Step size

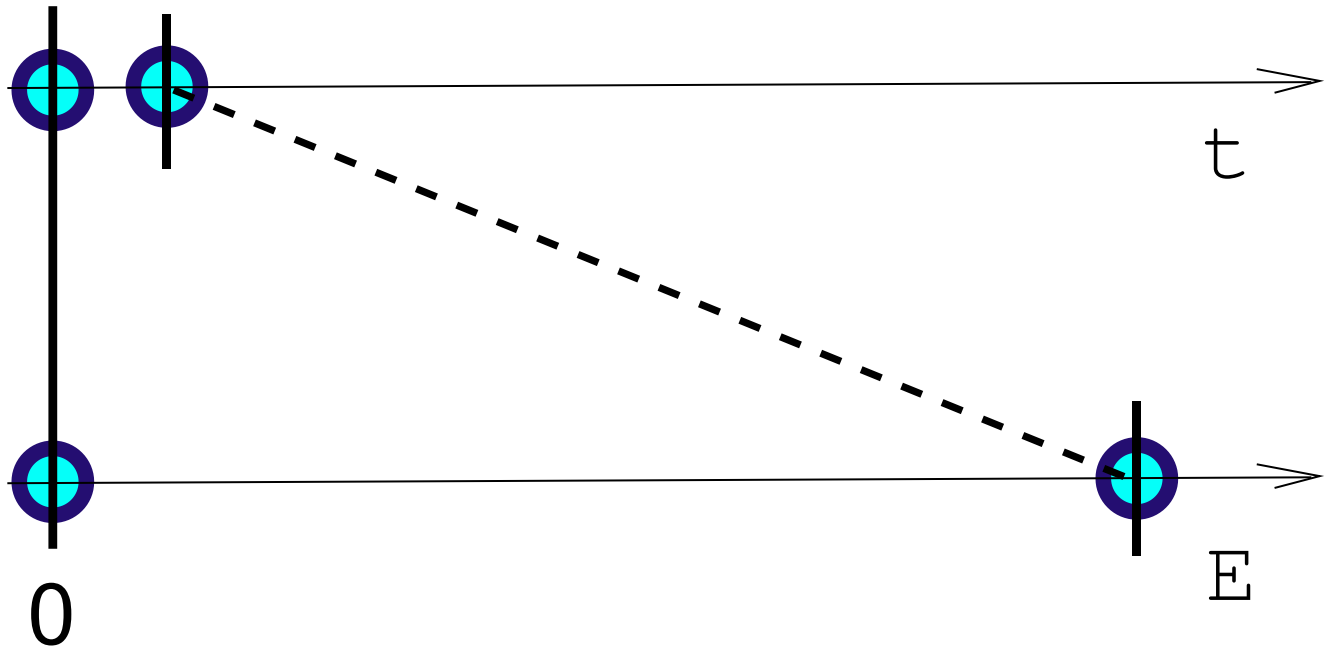


Molecular Dynamics: the boundary conditions

What about the total number of steps?

Time limits:

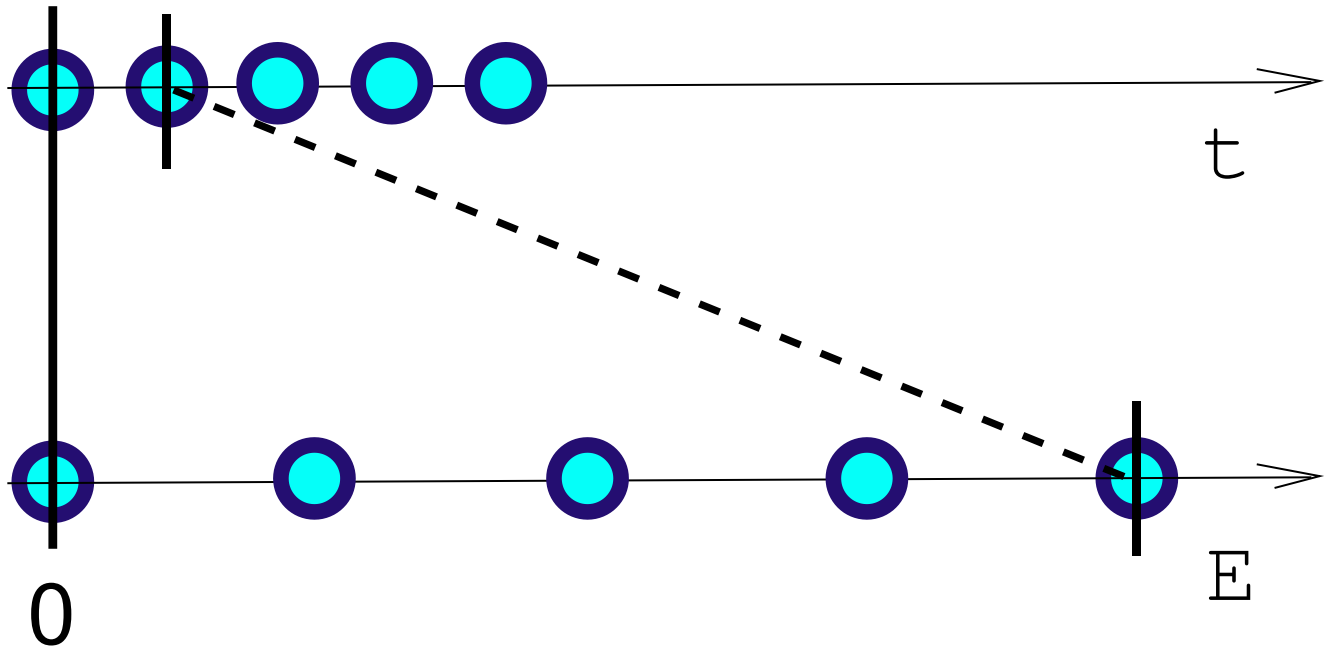
- Step size
- Number of steps



Molecular Dynamics: the boundary conditions

Time limits:

- Step size
- Number of steps

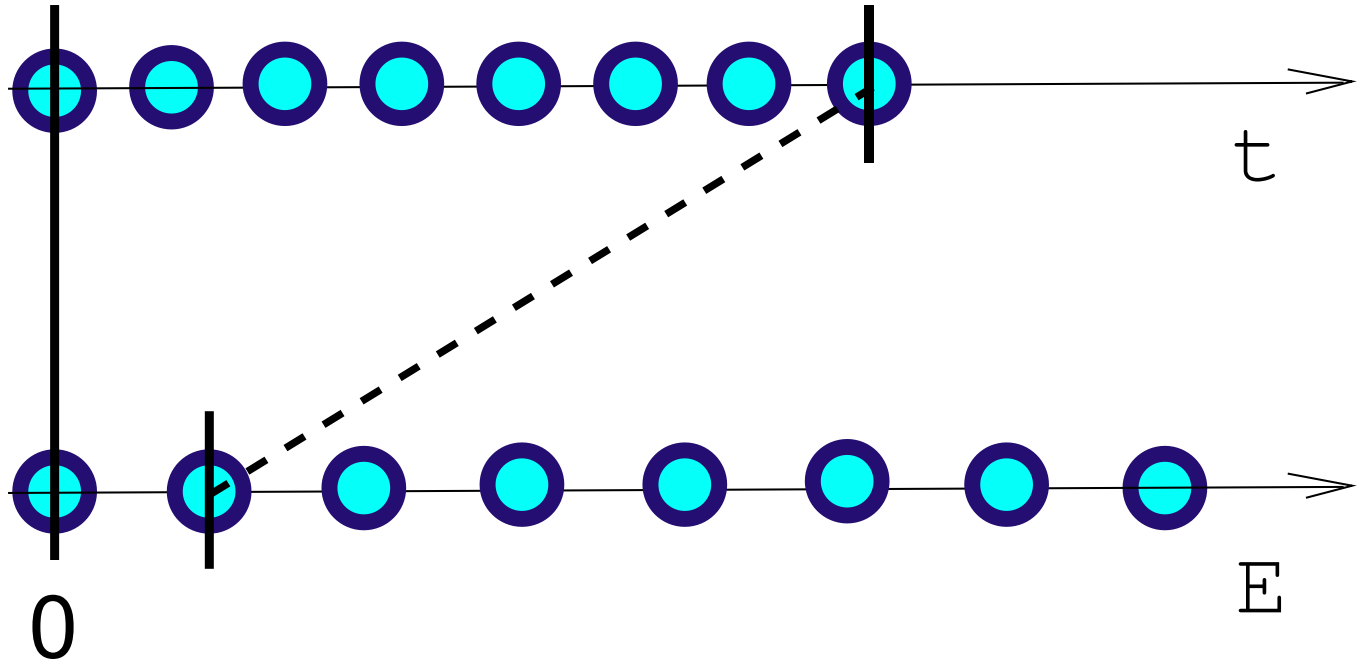


Molecular Dynamics: the boundary conditions

Adding more steps improves the sampling of the energy axis.

Time limits:

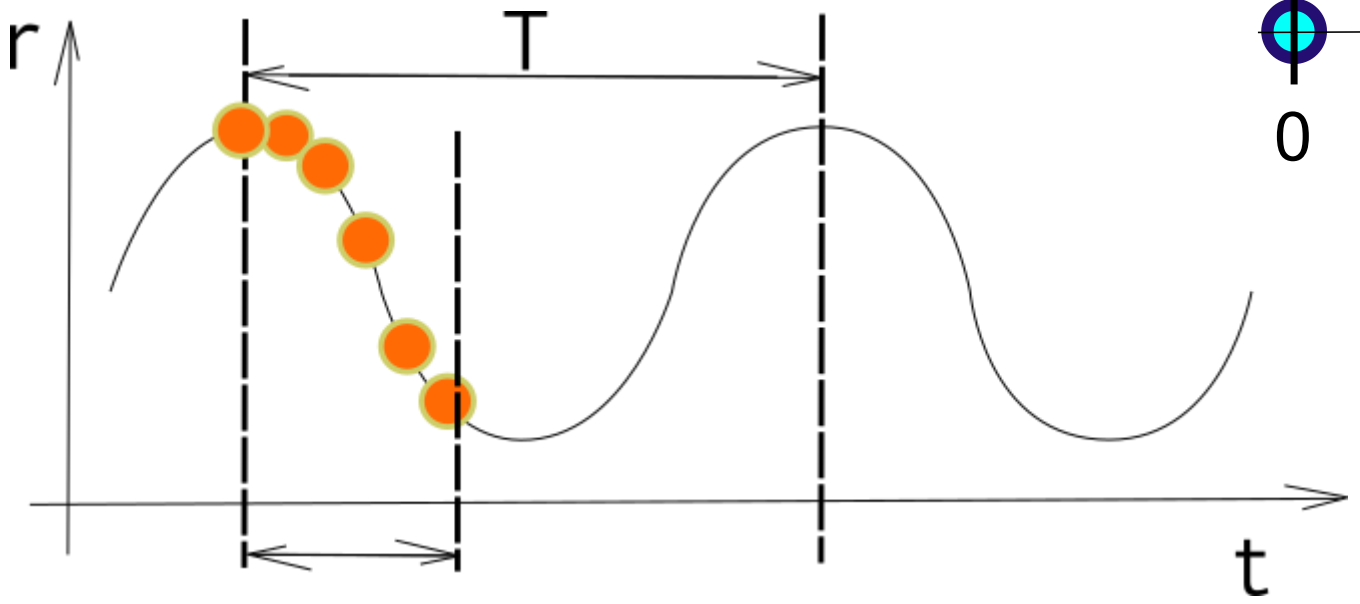
- Step size
- Number of steps

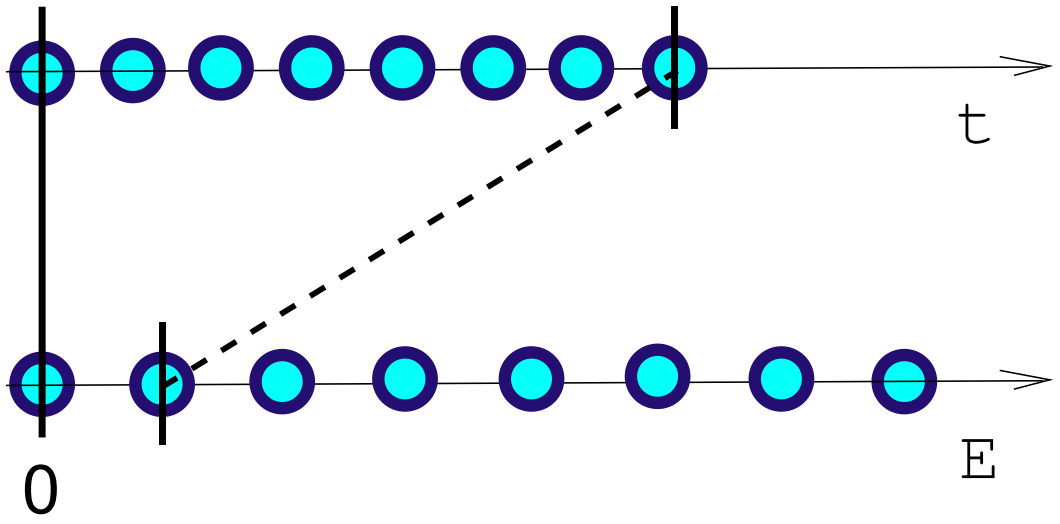


Molecular Dynamics: the boundary conditions

Slow oscillations:

Try to cover one full period or more.





Molecular Dynamics: the boundary conditions

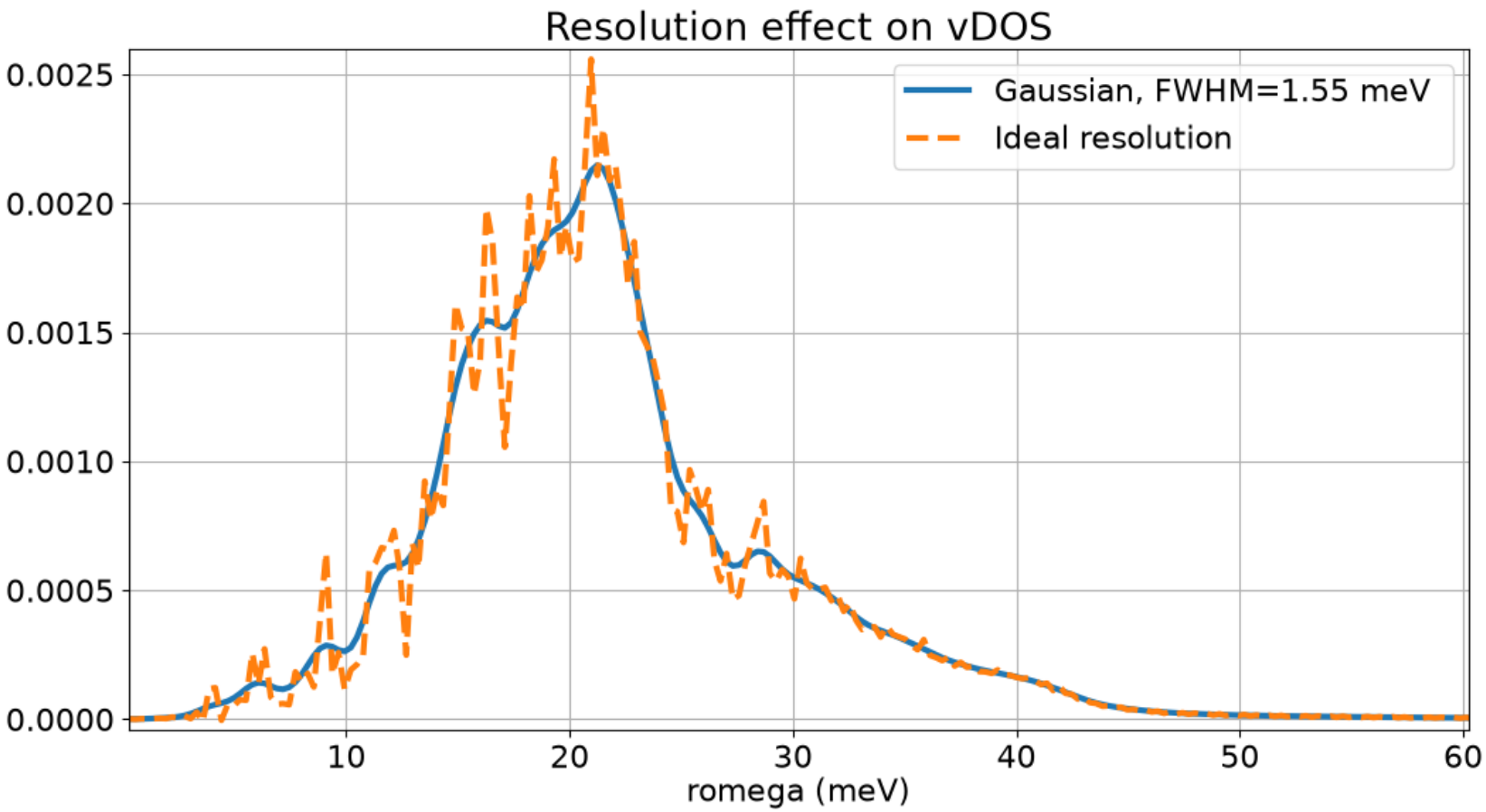
The time axis of the trajectory defines the time axis of the result.

Examples:

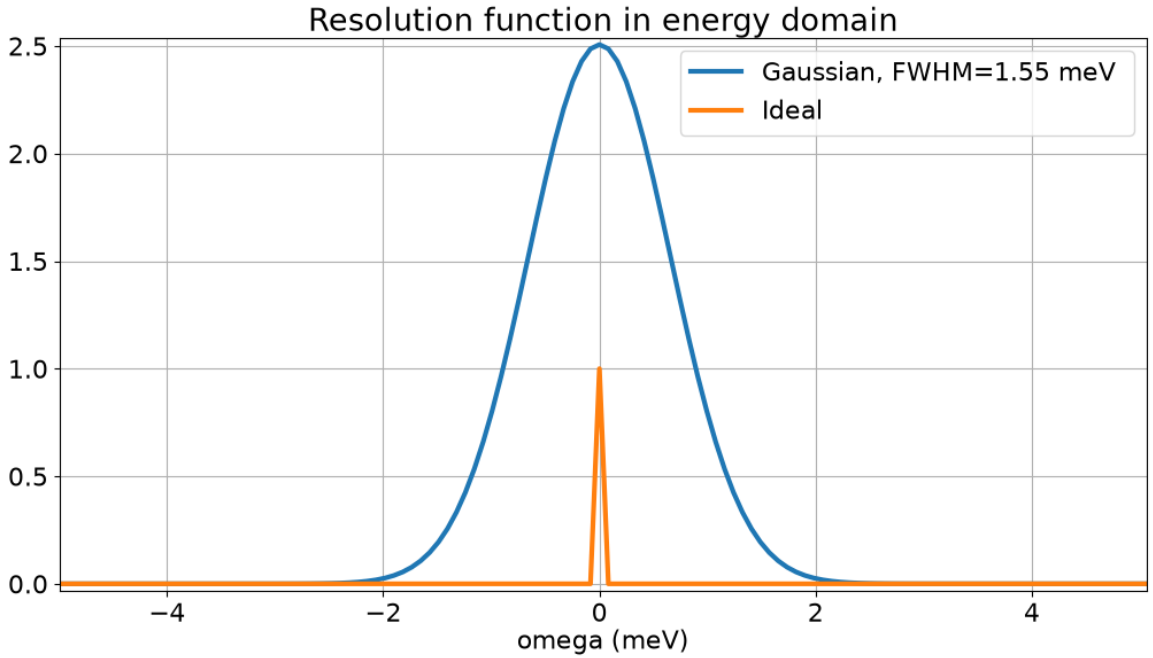
5 fs step x 100 frames = 500 fs total time => up to 412 meV in 4.1 meV steps
10 fs step x 50 frames = 500 fs total time => up to 206 meV in 4.1 meV steps

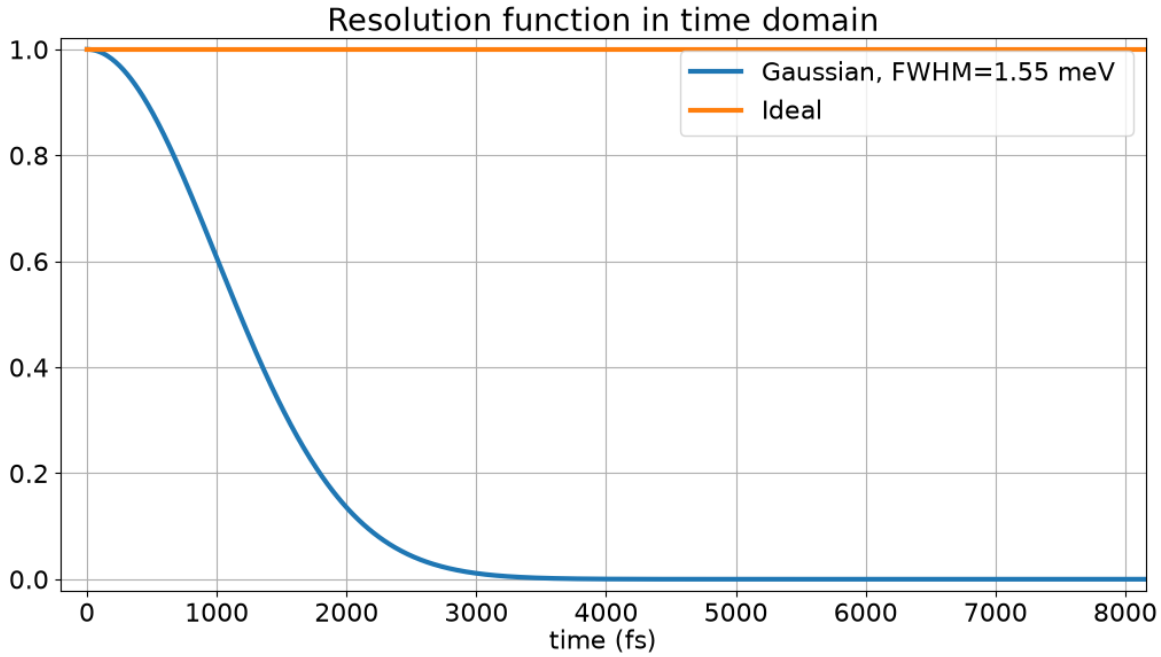
5 fs step x 10000 frames = 50 ps total time => up to 412 meV in 0.041 meV steps
1 ps step x 50 frames = 50 ps total time => up to 2.05 meV in 0.041 meV steps

Energy resolution

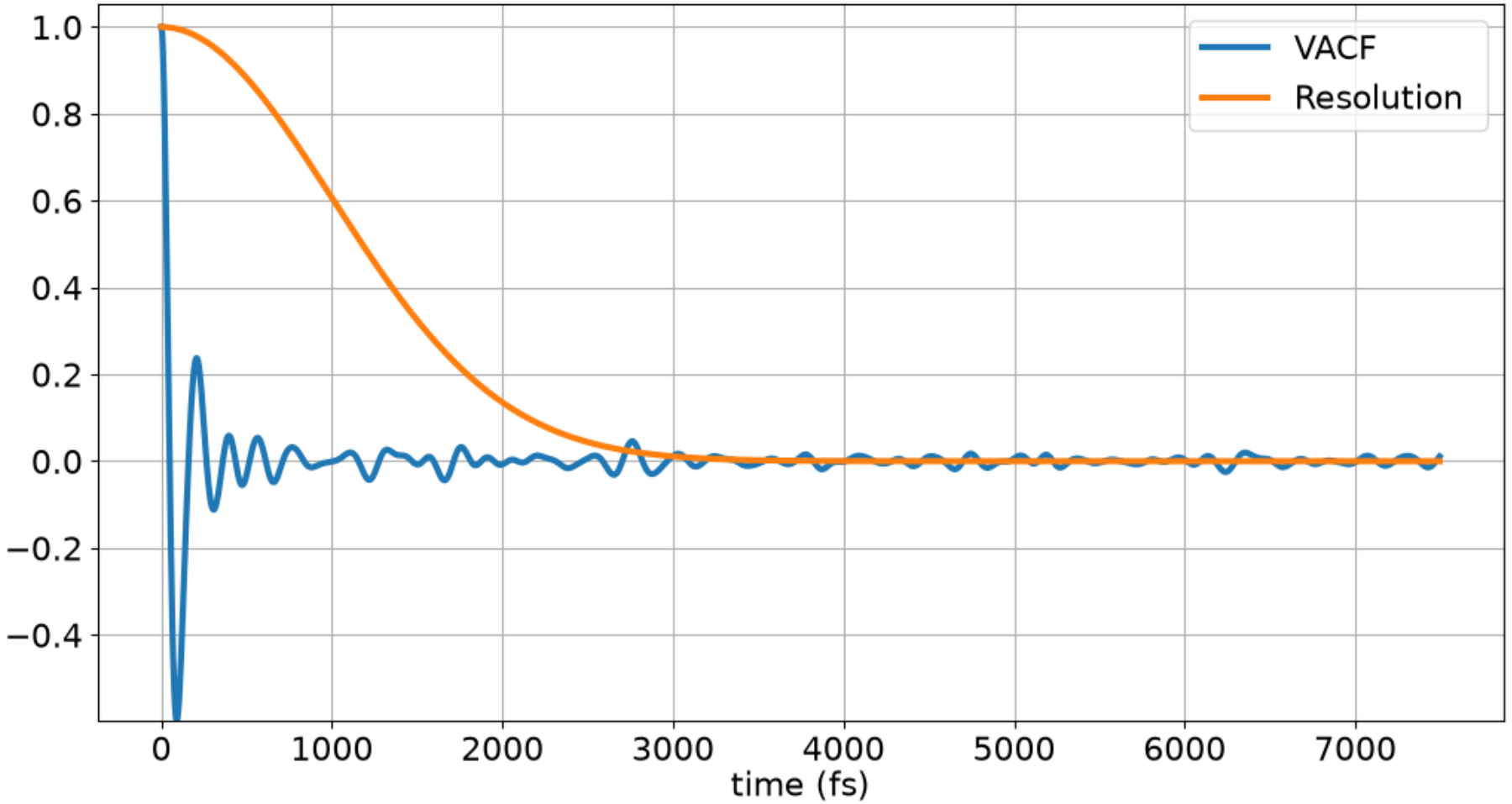


Energy resolution as a time window





Energy resolution as a time window



Neutrons: how are neutron results different?

https://www.ncnr.nist.gov/resources/n-lengths/

Neutron scattering lengths and cross sections							
Isotope	**conc**	**Coh b**	**Inc b**	**Coh xs**	**Inc xs**	**Scatt xs**	**Abs xs**
H	---	-3.7390	---	1.7568	80.26	82.02	0.3326
1H	99.985	-3.7406	25.274	1.7583	80.27	82.03	0.3326
2H	0.015	6.671	4.04	5.592	2.05	7.64	0.000519
3H	(12.32 a)	4.792	-1.04	2.89	0.14	3.03	0

We will talk more about coherent and incoherent scattering lengths in the tutorials on Wednesday.

Neutron scattering lengths can be used as **weights** when combining partial calculation results.

Neutrons: isotopes and contrast matching

MDANSE calculates Static Structure Factor as:

$$S_{\alpha\beta}(q) = 1 + \frac{4\pi\rho}{q}\int_0^\infty \mathrm{d}r\ [g_{\alpha\beta}(r) - 1]\, r \sin(qr)$$

$$g_{\alpha\beta}(r) = \frac{1}{N c_\alpha c_\beta}\frac{1}{4\pi r^2}\frac{1}{\rho}\sum_{j\in N_\alpha}\sum_{\substack{k\in N_\beta \\ k\neq j}}\langle\delta(r - |\mathbf{r}_k + \mathbf{r}_j|)\rangle$$

This is where **scattering lengths** become part of the calculation.

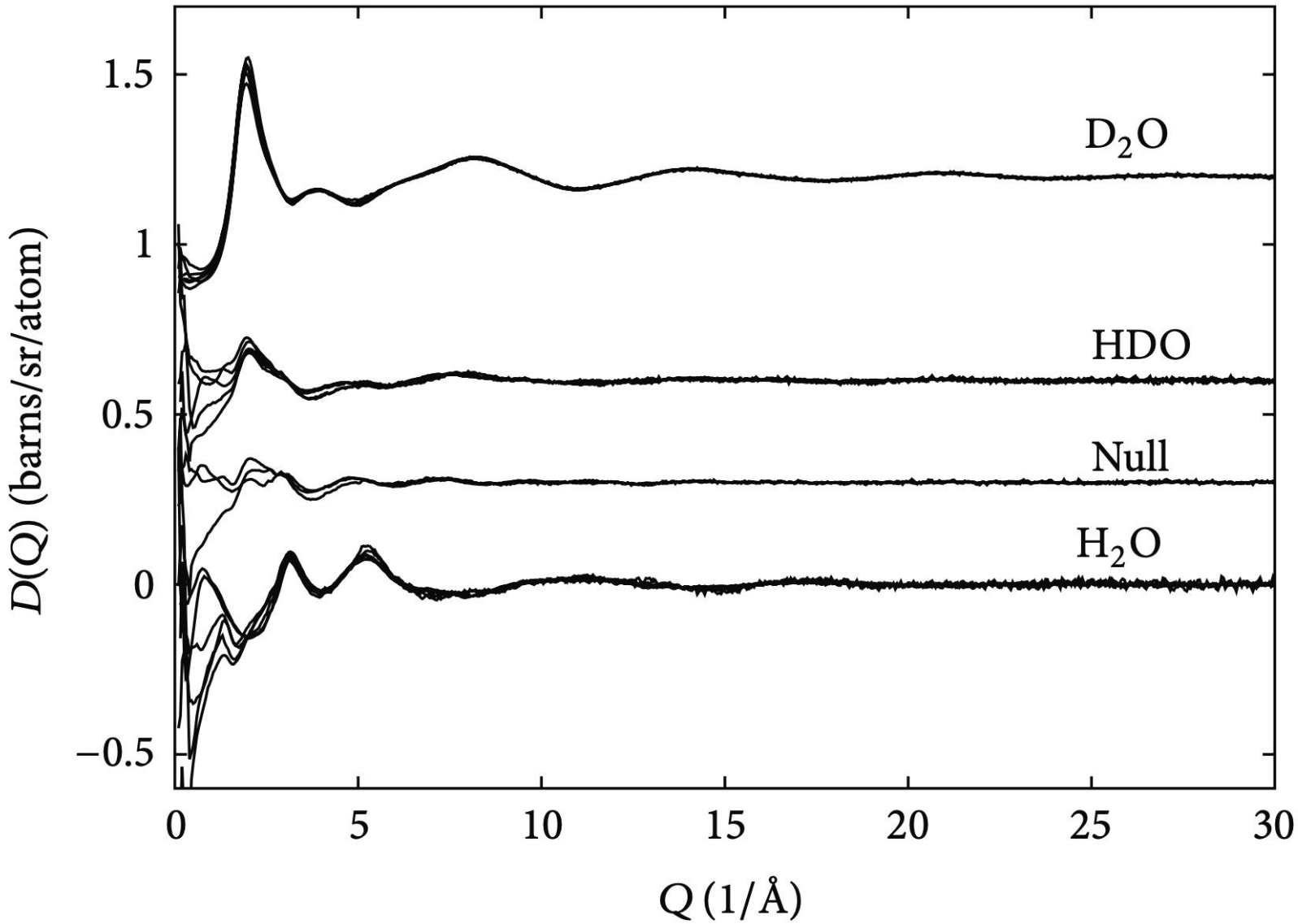
$$W_{\alpha\beta} = [2 - \delta_{\alpha\beta}]\frac{c_\alpha c_\beta\ \mathrm{Re}\left[b^\dagger_{\mathrm{coh},\alpha} b_{\mathrm{coh},\beta}\right]}{\sum_{\gamma\delta} c_\gamma c_\delta b^\dagger_{\mathrm{coh},\gamma} b_{\mathrm{coh},\delta}}$$

$$S(q) = \sum_{\alpha,\beta} W_{\alpha\beta} S_{\alpha\beta}(q)$$

Neutrons: isotopes and contrast matching

The Radial Distribution Functions of Water as Derived from Radiation Total Scattering Experiments:
Is There Anything We Can Say for Sure?

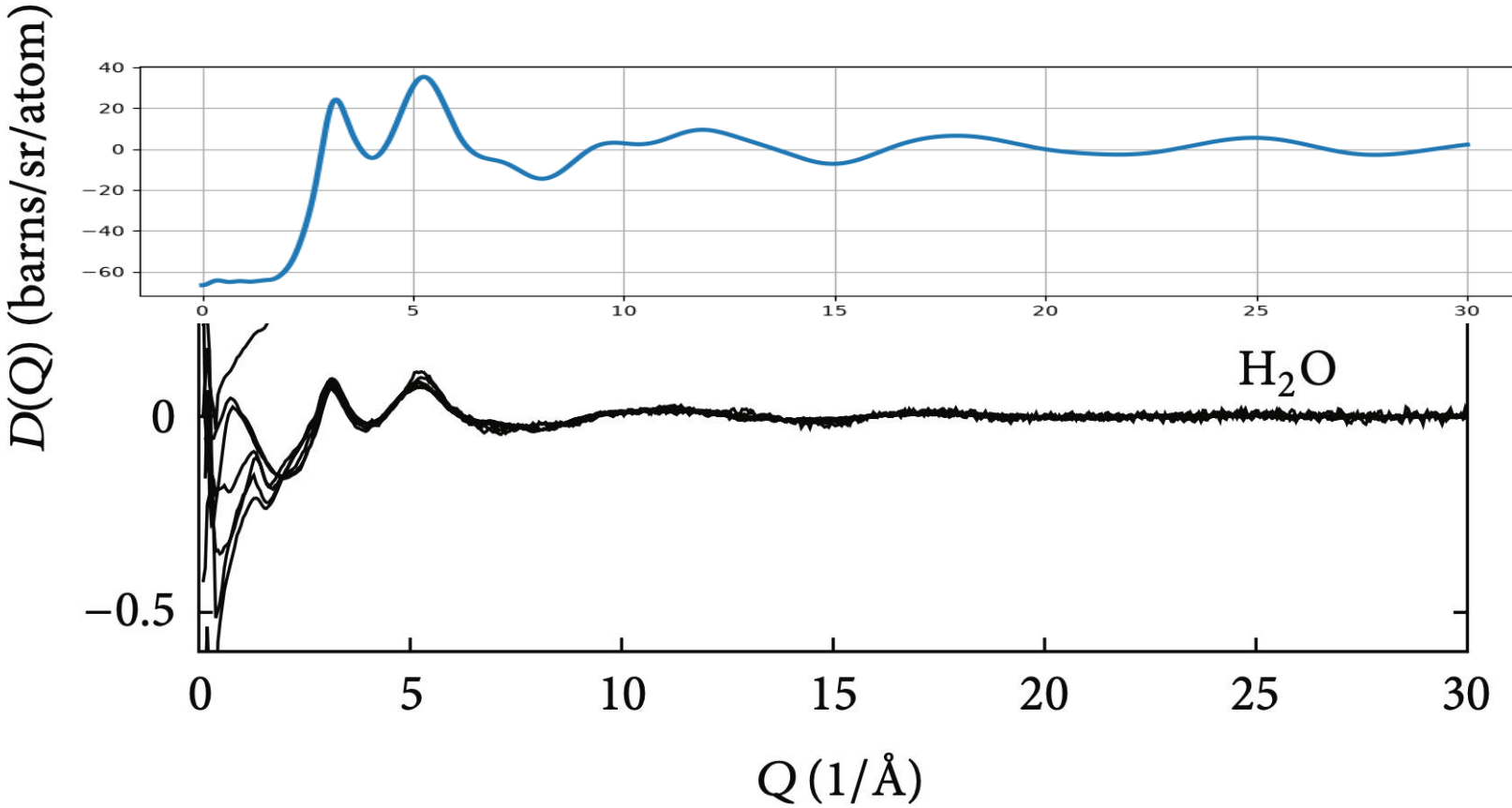
A. K. Soper
http://dx.doi.org/10.1155/2013/279463



Neutrons: isotopes and contrast matching

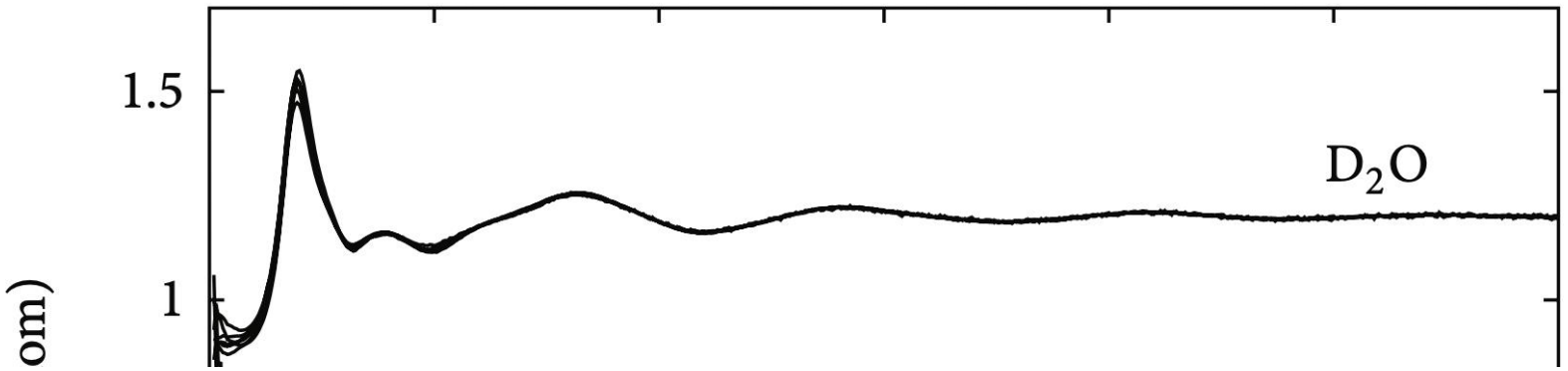
Static structure factor.
Coherent scattering length is the weight in the calculation.

Isotopically normal water.

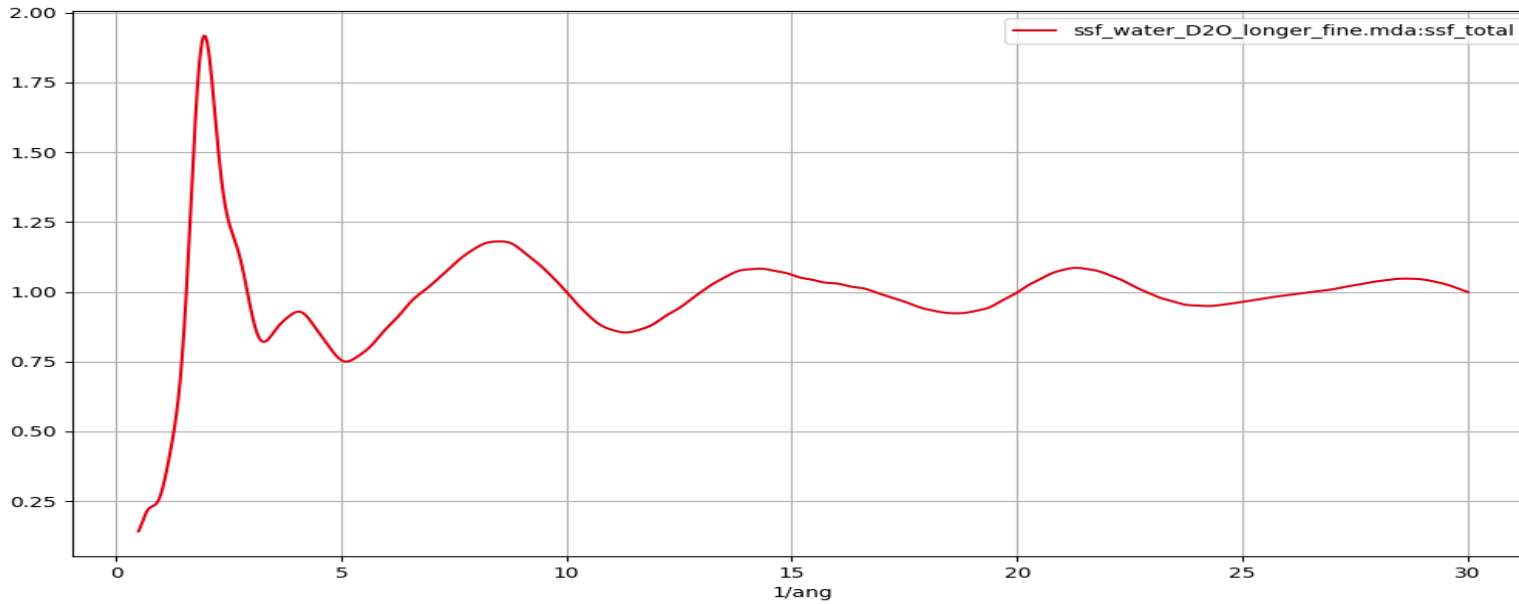


Neutrons: isotopes and contrast matching

Static structure factor.
Coherent scattering length is the weight in the calculation.



Fully deuterated water.



QENS: Dynamic Structure Factor

Typically, we calculate two components of the Dynamic Structure Factor:

1. Dynamic **Incoherent** Structure Factor
- proportional to the **incoherent** scattering cross section of contributing atoms,
- calculated from correlations of individual atom positions over time.

2. Dynamic **Coherent** Structure Factor
- proportional to the **coherent** scattering cross section of contributing atoms,
- calculated from correlations between different atoms over time,
- calculation on vectors commensurate with the reciprocal unit cell.

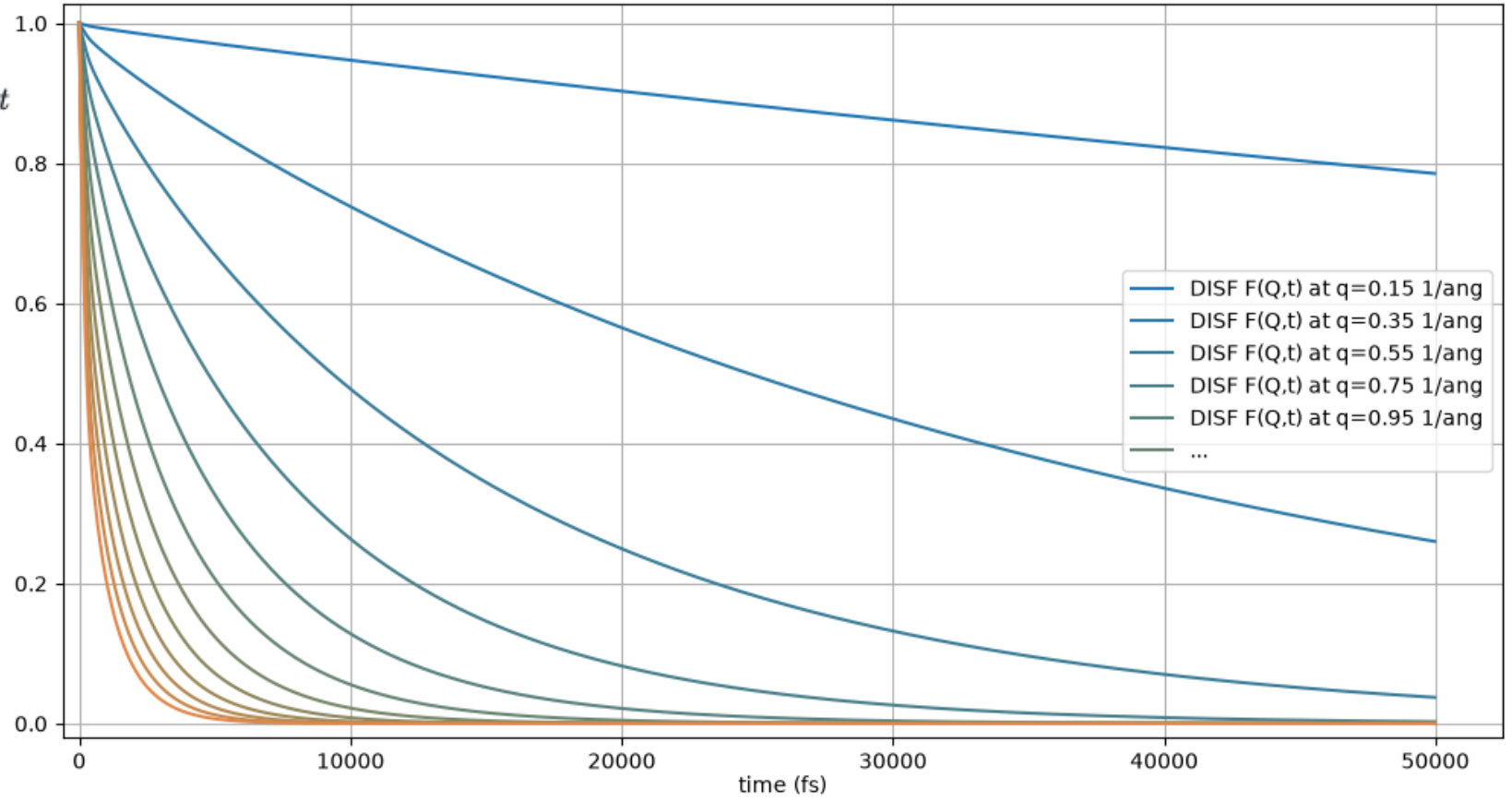
QENS: Dynamic Incoherent Structure Factor

$$F_{\mathrm{inc},\alpha}(\mathbf{q},t) = \frac{1}{Nc_\alpha}\sum_{j\in N_\alpha}\mathrm{Re}\left[\left\langle \exp\left[-i\mathbf{q}\cdot\mathbf{r}_j(0)\right]\exp\left[i\mathbf{q}\cdot\mathbf{r}_j(t)\right]\right\rangle\right]$$

$$S_{\mathrm{inc},\alpha}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{inc},\alpha}(\mathbf{q},t)e^{-i\omega t}$$

$$S_{\mathrm{inc}}(\mathbf{q},\omega) = \sum_\alpha W_\alpha S_{\mathrm{inc},\alpha}(\mathbf{q},\omega)$$

$$F_{\mathrm{inc}}(\mathbf{q},t) = \sum_\alpha W_\alpha F_{\mathrm{inc},\alpha}(\mathbf{q},t)$$



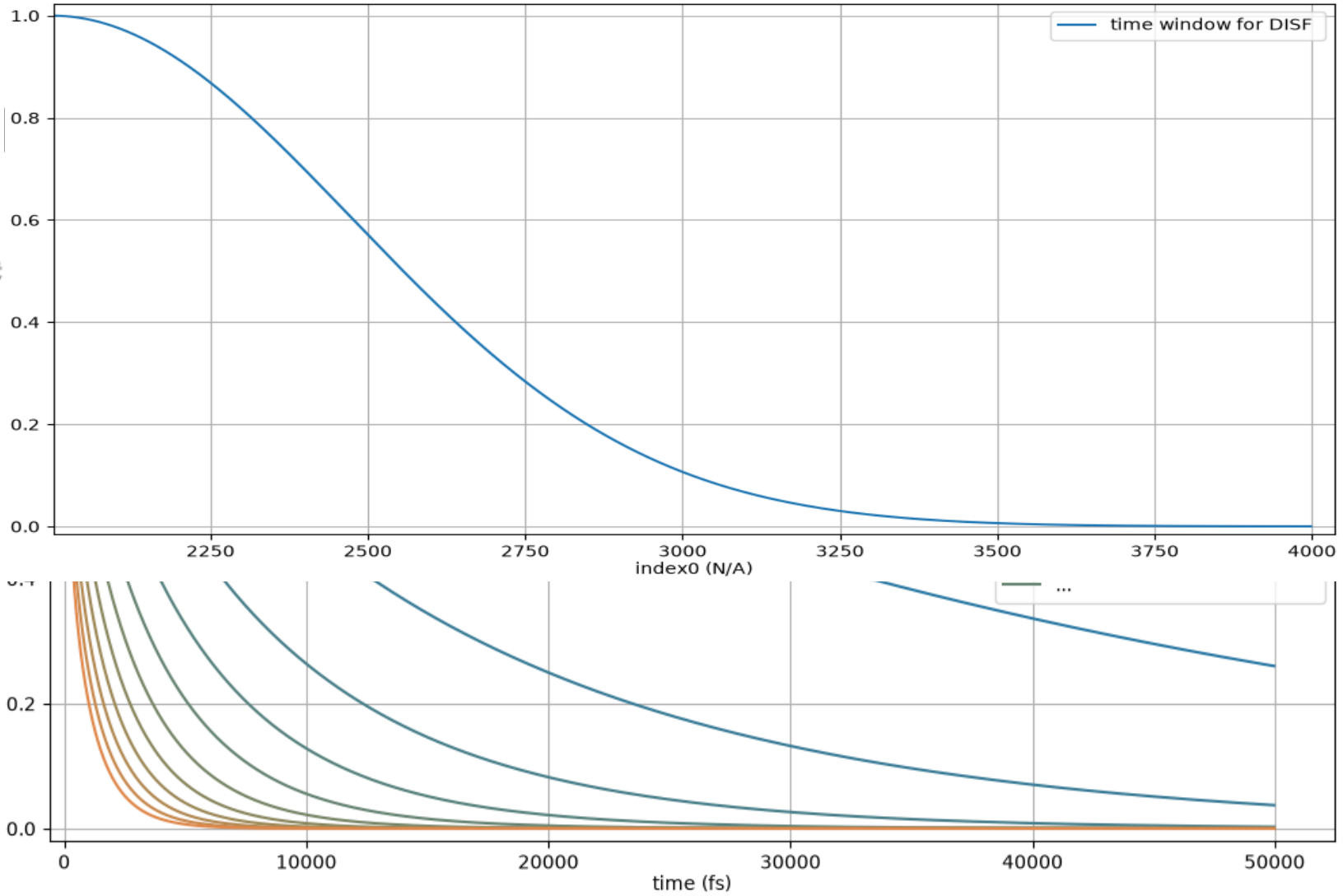
QENS: Dynamic Incoherent Structure Factor

$$F_{\mathrm{inc},\alpha}(\mathbf{q},t) = \frac{1}{Nc_\alpha}\sum_{j\in N_\alpha}\mathrm{Re}\left[\langle\exp\right.$$

$$S_{\mathrm{inc},\alpha}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{inc},\alpha}(\mathbf{q},t)e^{-i\omega t}$$

$$S_{\mathrm{inc}}(\mathbf{q},\omega) = \sum_\alpha W_\alpha S_{\mathrm{inc},\alpha}(\mathbf{q},\omega)$$

$$F_{\mathrm{inc}}(\mathbf{q},t) = \sum_\alpha W_\alpha F_{\mathrm{inc},\alpha}(\mathbf{q},t)$$



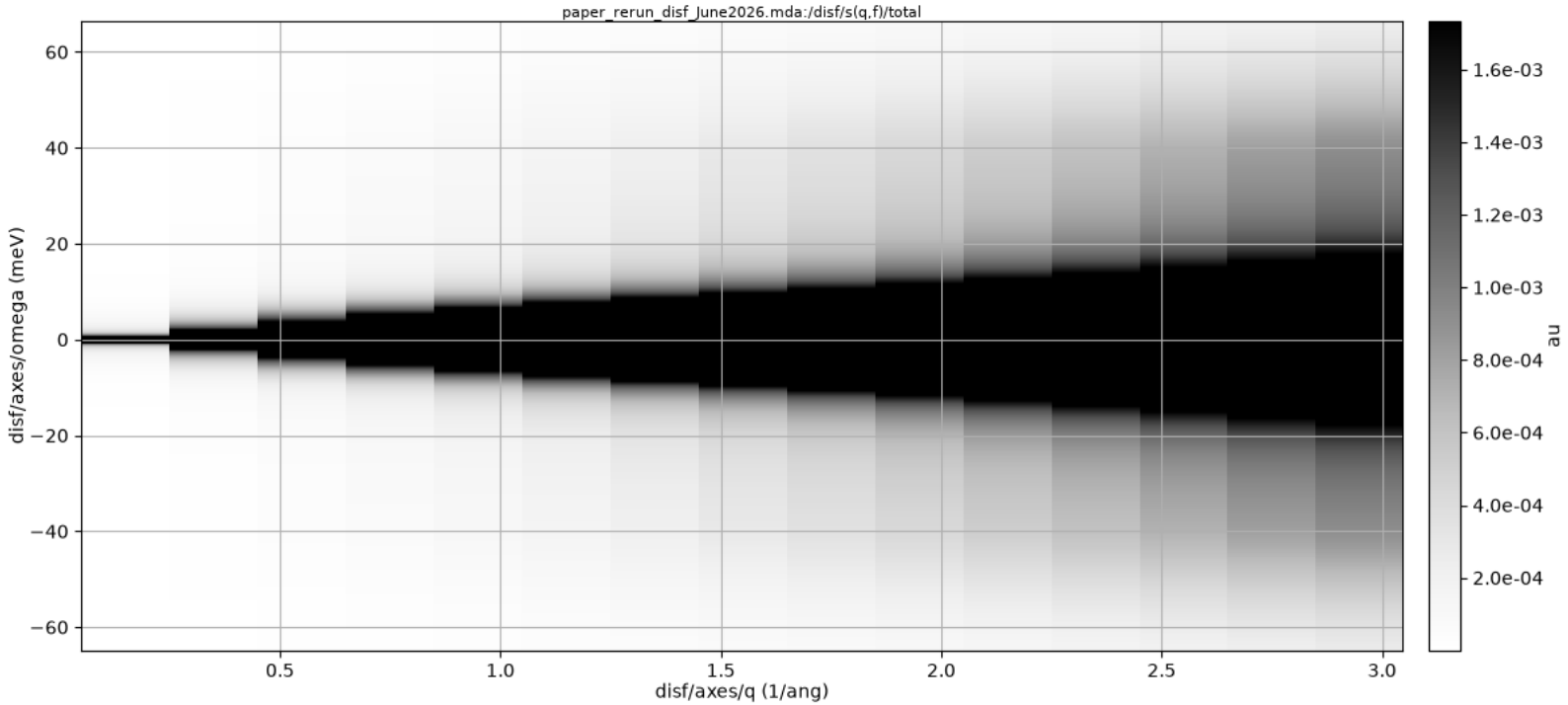
QENS: Dynamic Incoherent Structure Factor

$$F_{\mathrm{inc},\alpha}(\mathbf{q},t) = \frac{1}{Nc_\alpha}\sum_{j\in N_\alpha}\mathrm{Re}\left[\left\langle \exp\left[-i\mathbf{q}\cdot\mathbf{r}_j(0)\right]\exp\left[i\mathbf{q}\cdot\mathbf{r}_j(t)\right]\right\rangle\right]$$

$$S_{\mathrm{inc},\alpha}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{inc},\alpha}(\mathbf{q},t)e^{-i\omega t}$$

$$S_{\mathrm{inc}}(\mathbf{q},\omega) = \sum_\alpha W_\alpha S_{\mathrm{inc},\alpha}(\mathbf{q},\omega)$$

$$F_{\mathrm{inc}}(\mathbf{q},t) = \sum_\alpha W_\alpha F_{\mathrm{inc},\alpha}(\mathbf{q},t)$$



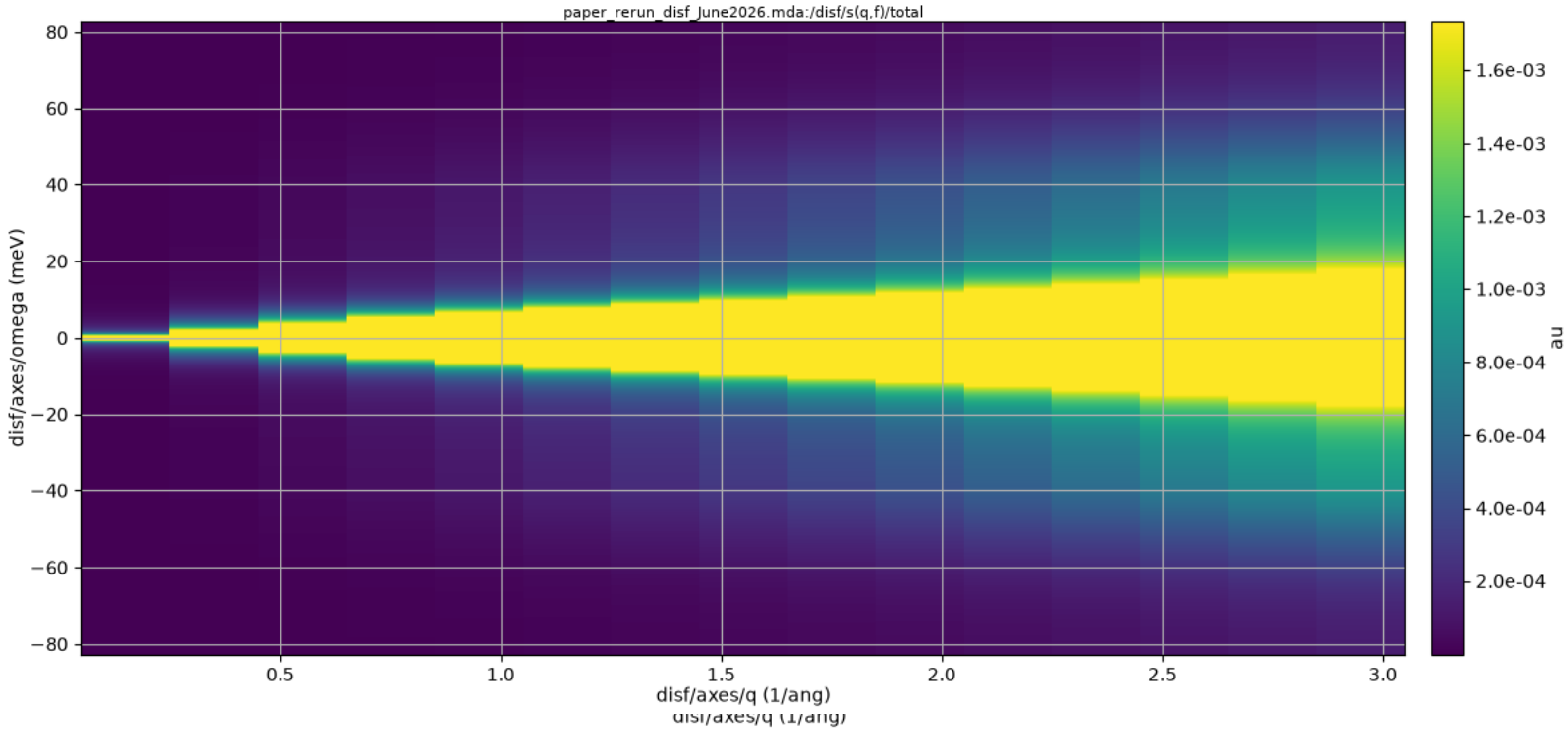
QENS: Dynamic Incoherent Structure Factor

$$F_{\mathrm{inc},\alpha}(\mathbf{q},t) = \frac{1}{Nc_\alpha}\sum_{j\in N_\alpha}\mathrm{Re}\left[\left\langle \exp\left[-i\mathbf{q}\cdot\mathbf{r}_j(0)\right]\exp\left[i\mathbf{q}\cdot\mathbf{r}_j(t)\right]\right\rangle\right]$$

$$S_{\mathrm{inc},\alpha}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{inc},\alpha}(\mathbf{q},t)e^{-i\omega t}$$

$$S_{\mathrm{inc}}(\mathbf{q},\omega) = \sum_\alpha W_\alpha S_{\mathrm{inc},\alpha}(\mathbf{q},\omega)$$

$$F_{\mathrm{inc}}(\mathbf{q},t) = \sum_\alpha W_\alpha F_{\mathrm{inc},\alpha}(\mathbf{q},t)$$



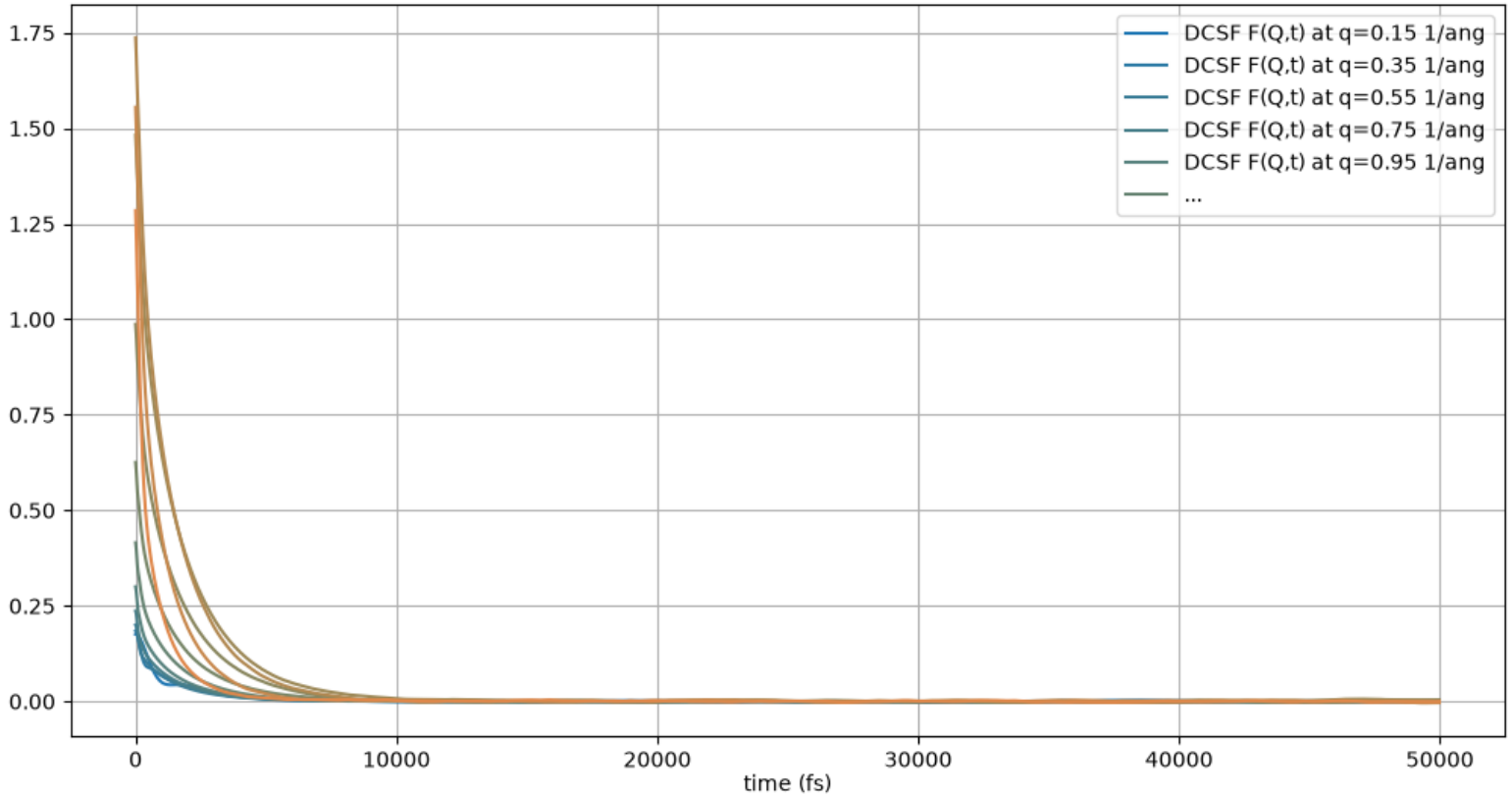
QENS: Dynamic Coherent Structure Factor

$$F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t) = \frac{1}{N\sqrt{c_\alpha c_\beta}} \sum_{j\in N_\alpha} \sum_{k\in N_\beta} \mathrm{Re}\left[\left\langle \exp\left[-i\mathbf{q}\cdot\mathbf{r}_j(0)\right] \exp\left[i\mathbf{q}\cdot\mathbf{r}_k(t)\right]\right\rangle\right]$$

$$S_{\mathrm{coh},\alpha\beta}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t) e^{-i\omega t}$$

$$F_{\mathrm{coh}}(\mathbf{q},t) = \sum_\alpha \sum_{\beta\geq\alpha} W_{\alpha\beta} F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t)$$

$$S_{\mathrm{coh}}(\mathbf{q},\omega) = \sum_\alpha \sum_{\beta\geq\alpha} W_{\alpha\beta} S_{\mathrm{coh},\alpha\beta}(\mathbf{q},\omega)$$



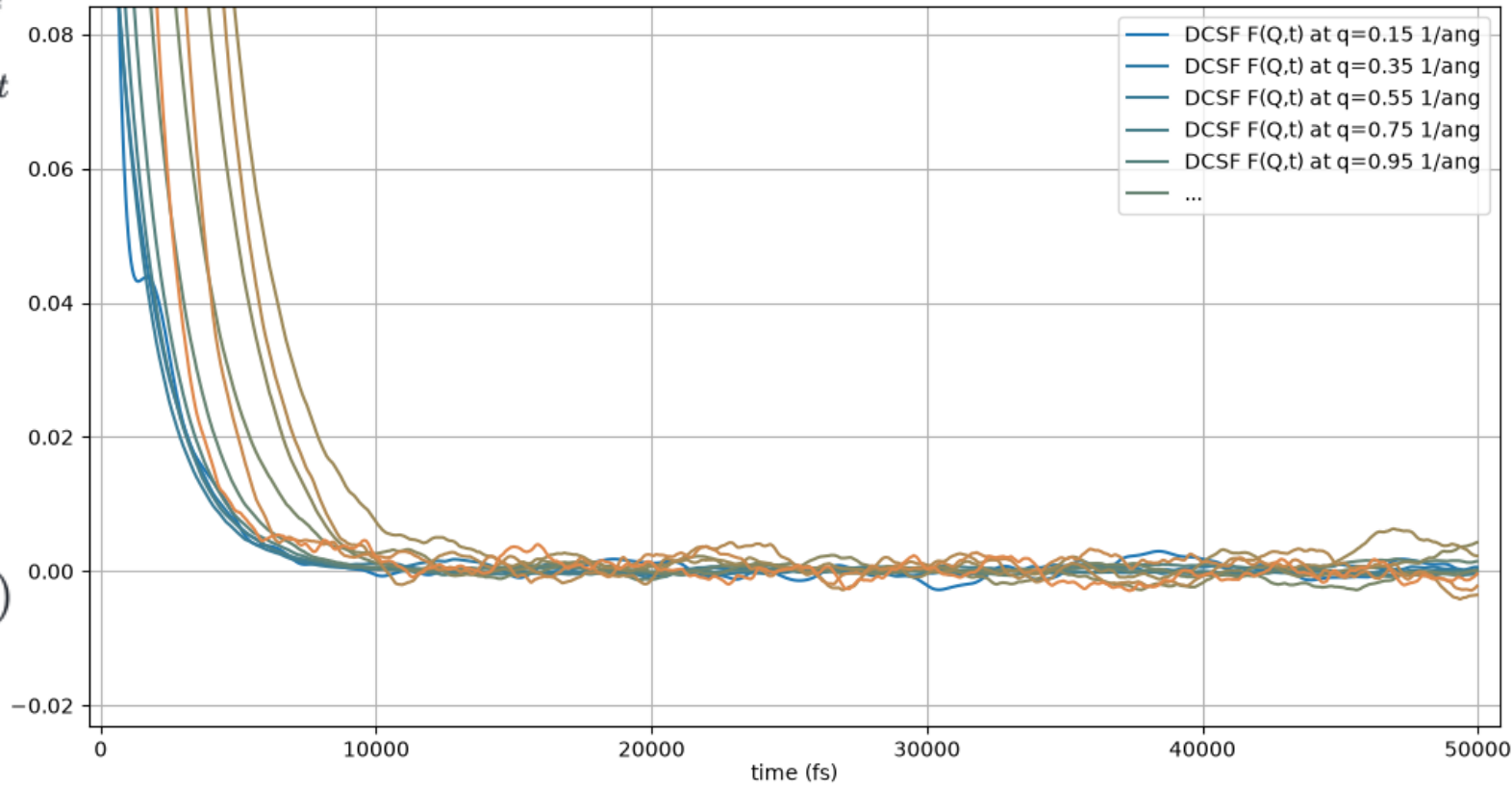
QENS: Dynamic Coherent Structure Factor

$$F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t) = \frac{1}{N\sqrt{c_\alpha c_\beta}} \sum_{j\in N_\alpha} \sum_{k\in N_\beta} \mathrm{Re}\left[\langle \exp\left[-i\mathbf{q}\cdot\mathbf{r}_j(0)\right] \exp\left[i\mathbf{q}\cdot\mathbf{r}_k(t)\right]\rangle\right]$$

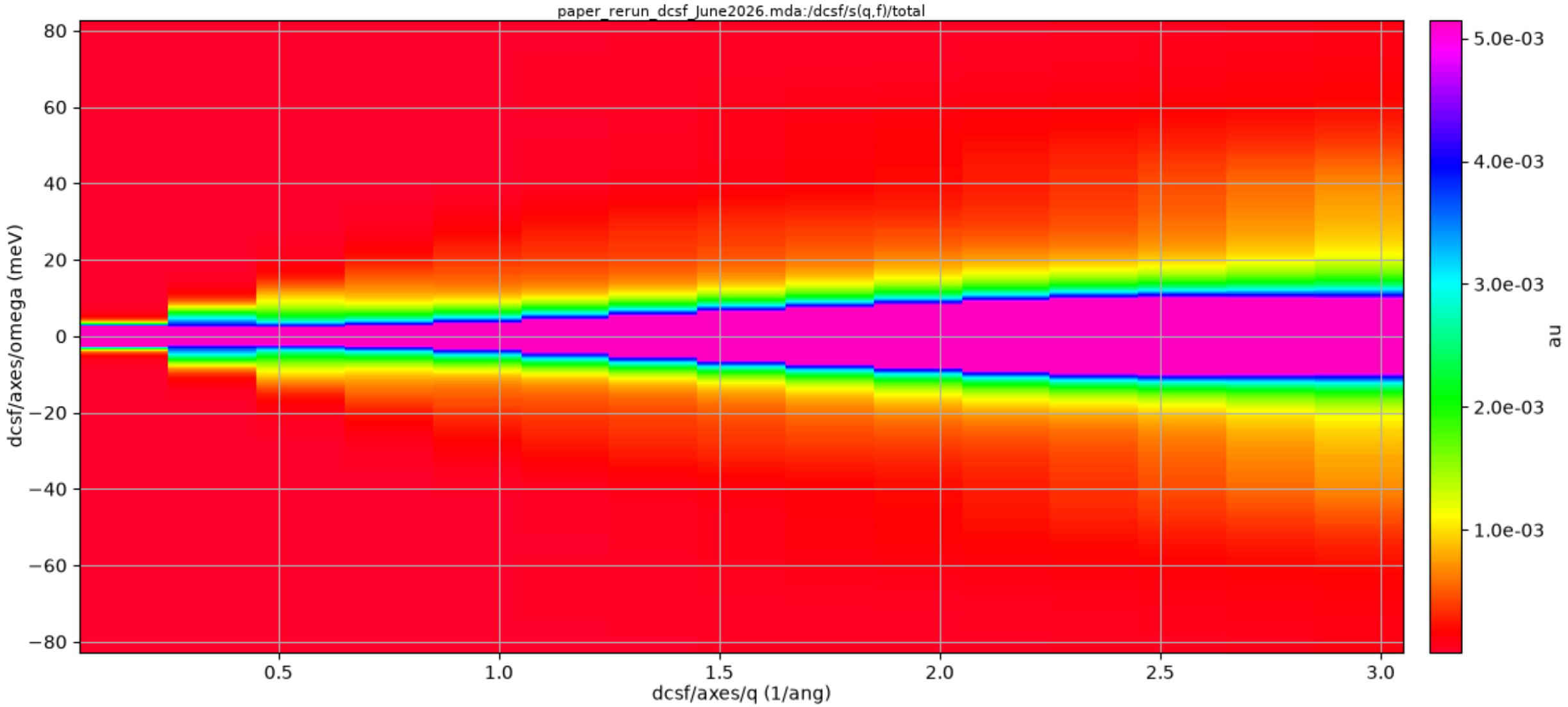
$$S_{\mathrm{coh},\alpha\beta}(\mathbf{q},\omega) = \int \mathrm{d}t\, F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t) e^{-i\omega t}$$

$$F_{\mathrm{coh}}(\mathbf{q},t) = \sum_\alpha \sum_{\beta\geq\alpha} W_{\alpha\beta} F_{\mathrm{coh},\alpha\beta}(\mathbf{q},t)$$

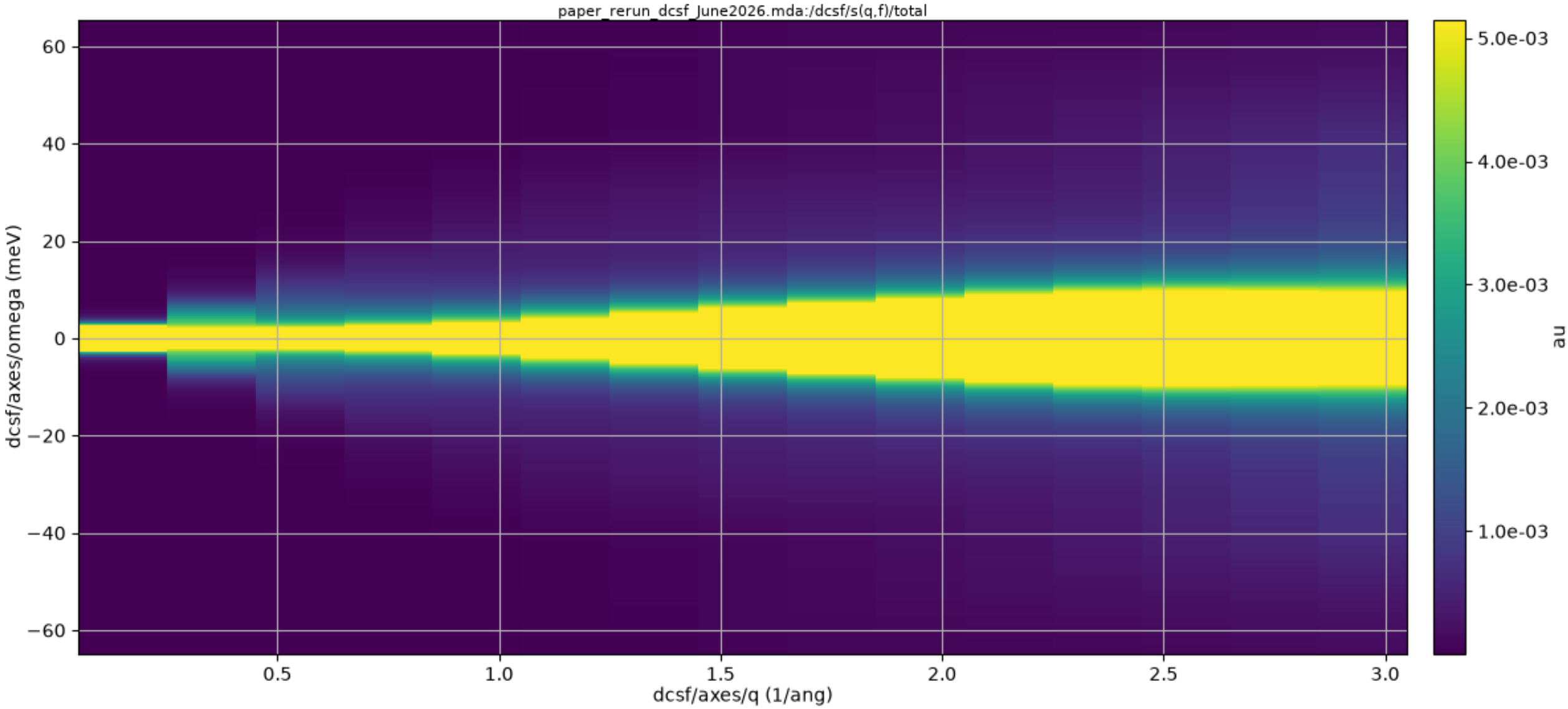
$$S_{\mathrm{coh}}(\mathbf{q},\omega) = \sum_\alpha \sum_{\beta\geq\alpha} W_{\alpha\beta} S_{\mathrm{coh},\alpha\beta}(\mathbf{q},\omega)$$



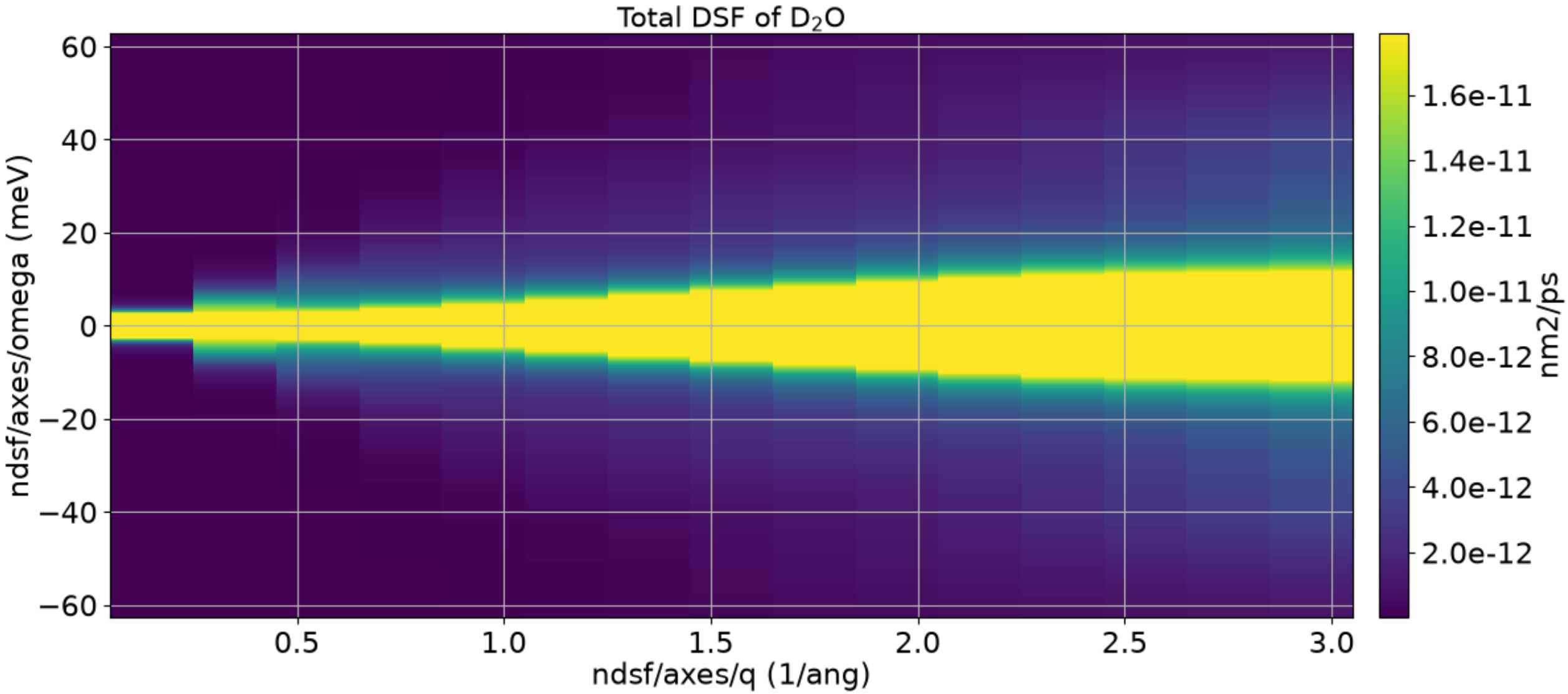
QENS: Dynamic Coherent Structure Factor



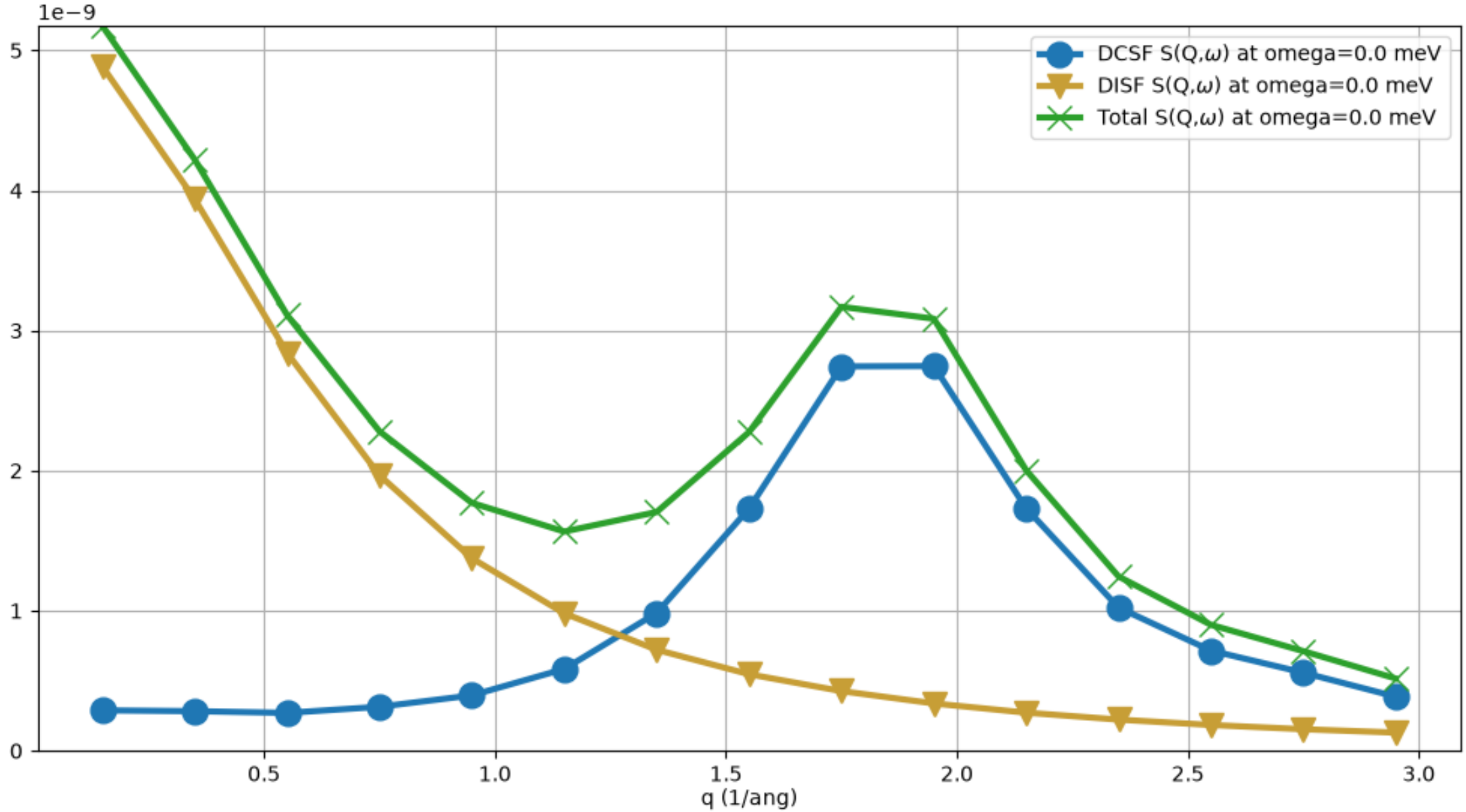
QENS: Dynamic Coherent Structure Factor



QENS: Total Dynamic Structure Factor

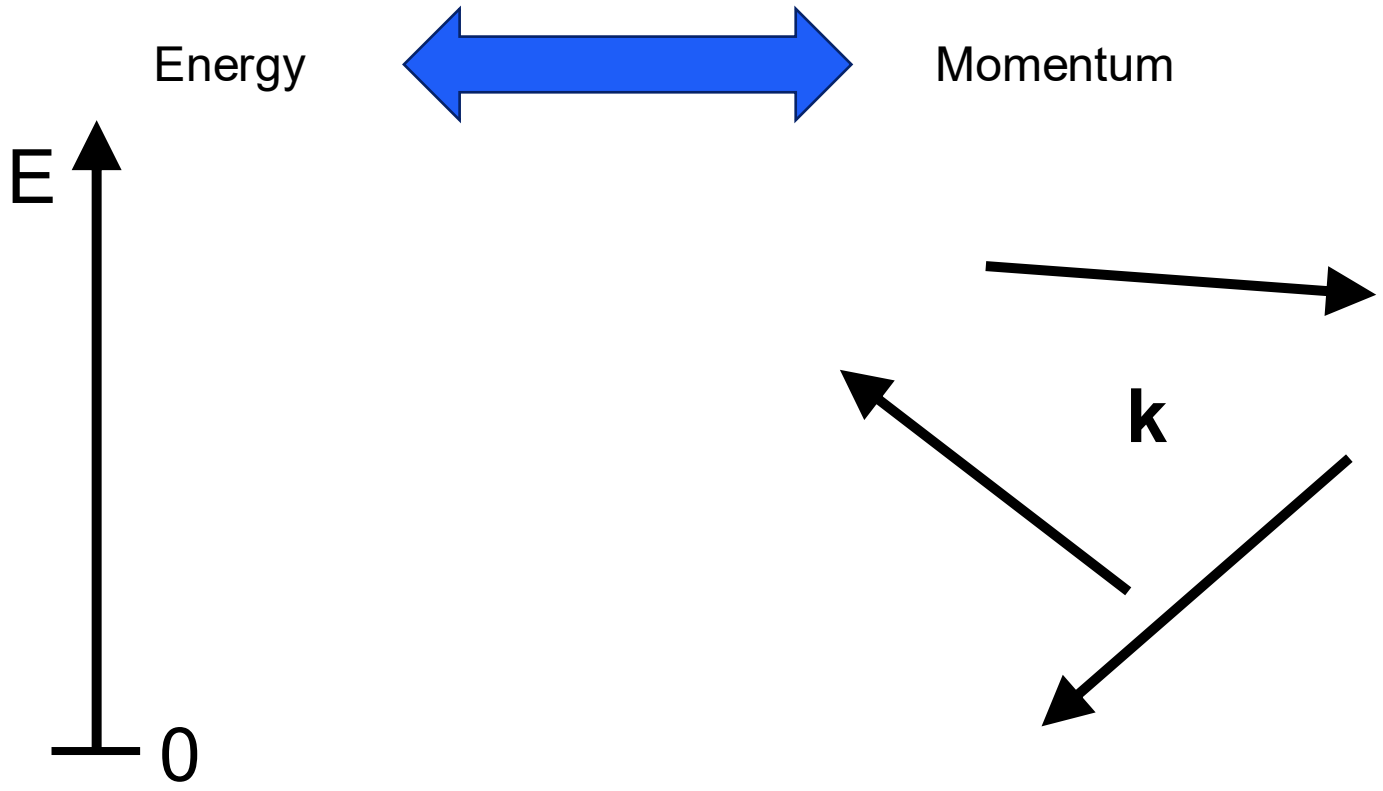


QENS: Total Dynamic Structure Factor



ISIS Ne
Muon S

Interlude: neutron properties



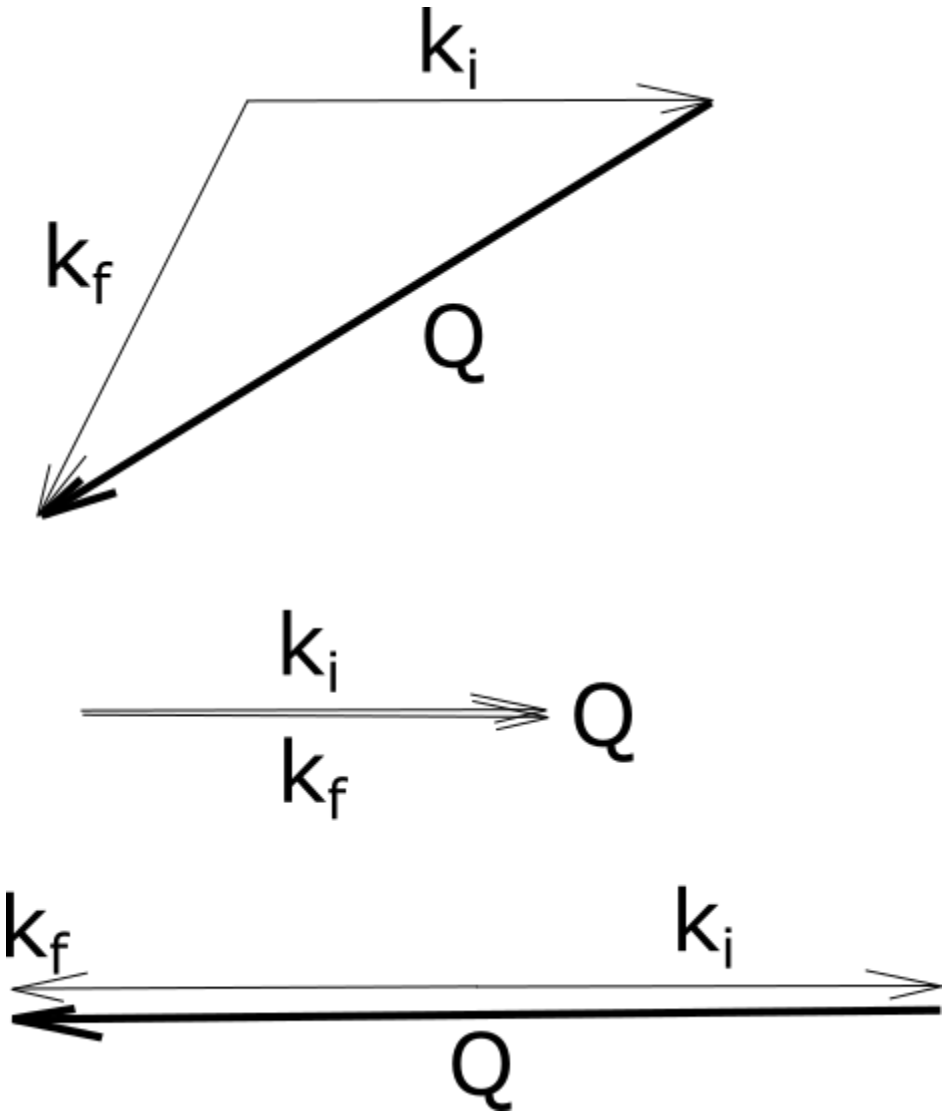
Neutrons: what is a Q vector?

Elastic scattering

$$|k_i| = |k_f|$$

For a single incoming neutron wavelength/energy/k vector

change of scattering direction changes the **direction** and **length** of **Q**



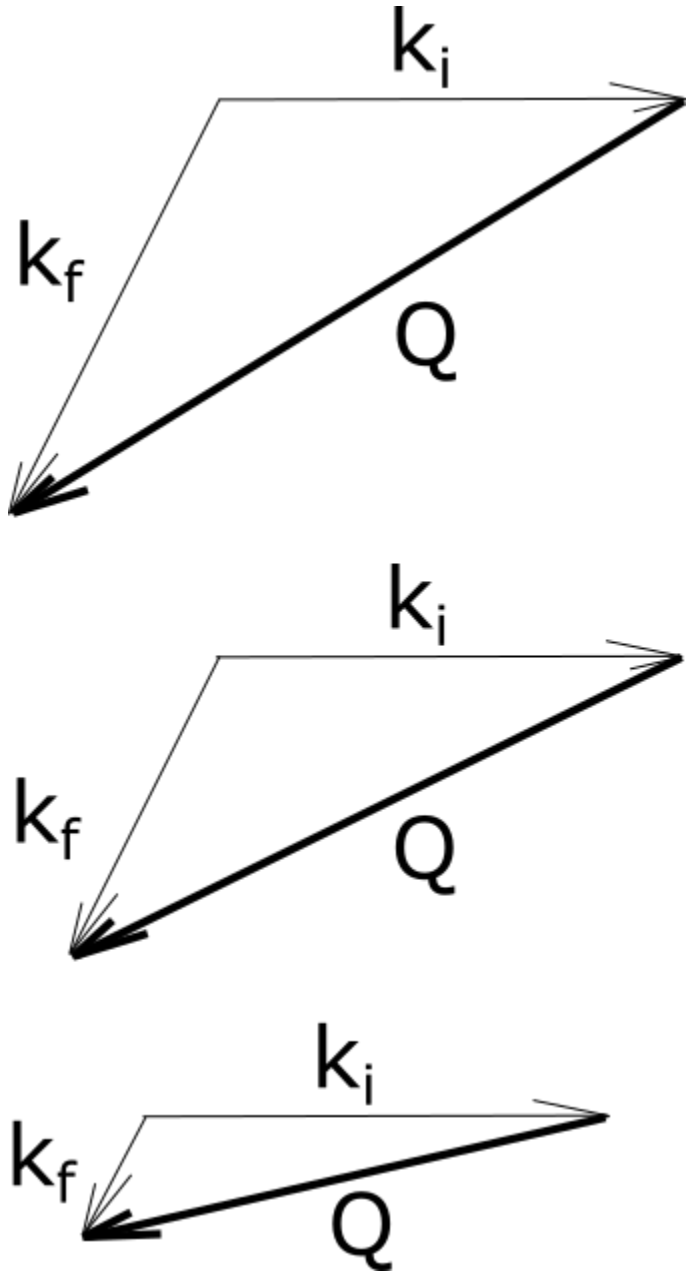
Neutrons: what is a Q vector?

Inelastic scattering

$$|k_i| \neq |k_f|$$

For a single incoming neutron wavelength/energy/k vector

a single scattering direction covers many **directions** and **lengths** of **Q**



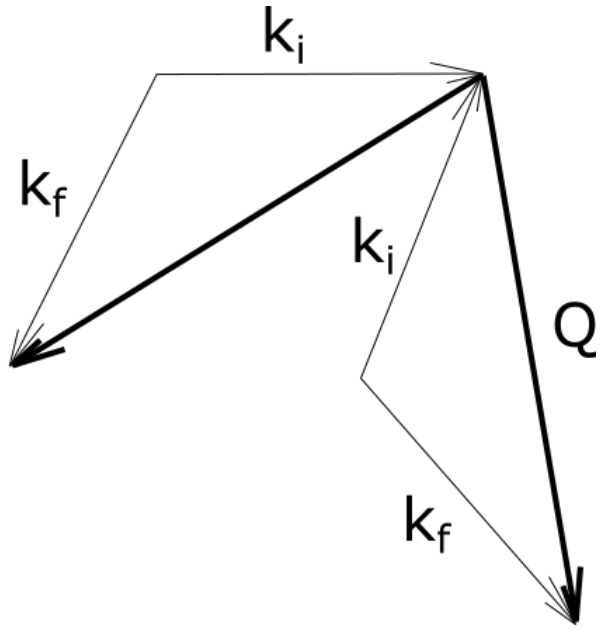
Neutrons: what is Q vector sampling?

In MDANSE:

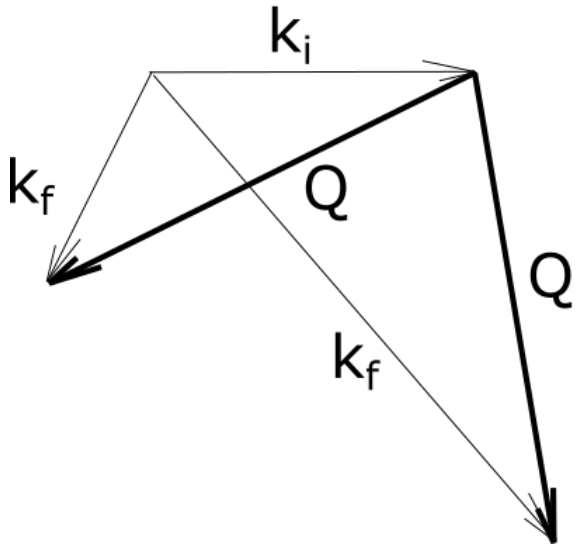
1. We generate Q-vector shells,
2. Results are averaged over all the vectors in a shell,
3. For each shell, the entire energy transfer range is covered.

In real life:

1. The $\mathbf{k_i}$ direction is fixed,
2. Not all $\mathbf{k_f}$ directions are possible,
3. Not for every **|Q|** value the entire energy range can be covered.



Elastic scattering



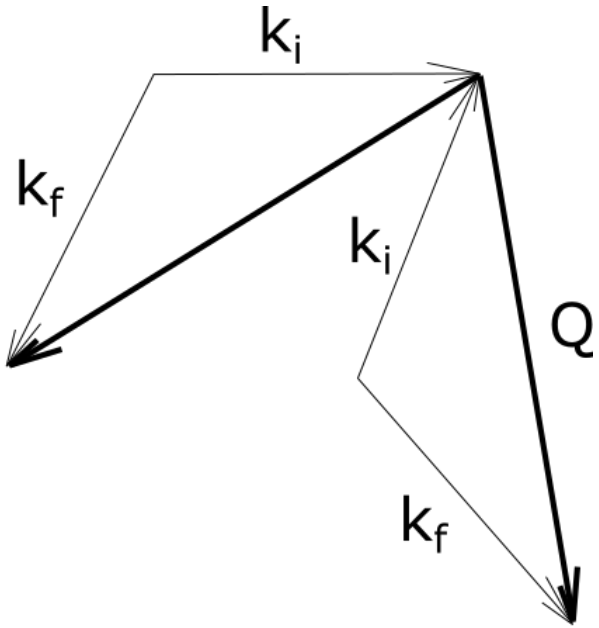
Inelastic scattering

Neutrons: what is Q vector sampling?

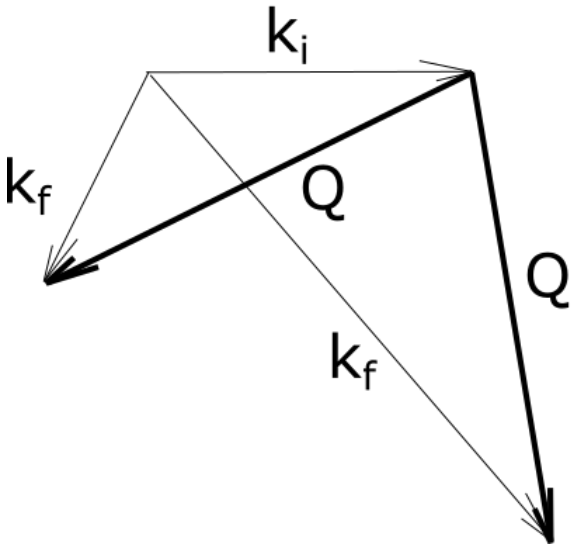
The QENS picture:

1. Energy range is small, so even for short Q vectors we should be able to cover it,
2. Energy range is small, so the resolution will be close to constant in the covered range.

Why do we generate multiple **Q** vectors with one **|Q|**?



Elastic scattering



Inelastic scattering

Neutrons: what is Q vector sampling?

Why do we generate multiple **Q** vectors with one **|Q|**?

Imagine you have a 0.01 mL sample of water.
The number of water molecules in this sample will be of the order of 10^{20}.

How does your trajectory compare to this?

Neutrons: what is Q vector sampling?

Why do we generate multiple **Q** vectors with one **|Q|**?

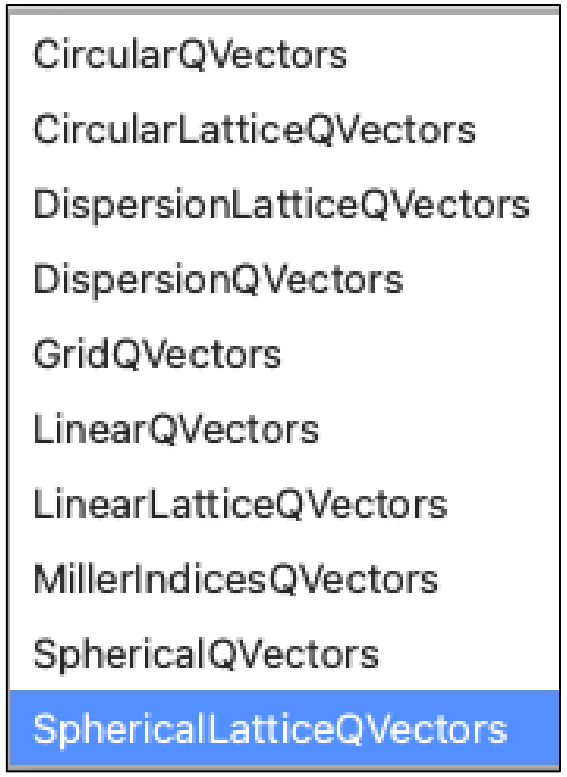
Imagine you have a 0.01 mL sample of water.
The number of water molecules in this sample will be of the order of 10^{20}.

How does your trajectory compare to this?

We don't sample all that many configurations in an MD trajectory.
Spherical sampling helps us re-use the same configurations in **different orientations**.

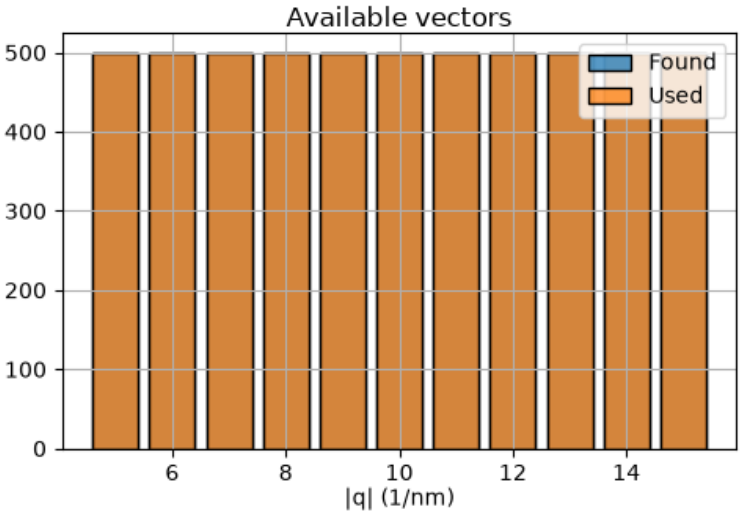
Do we need spherical averaging?

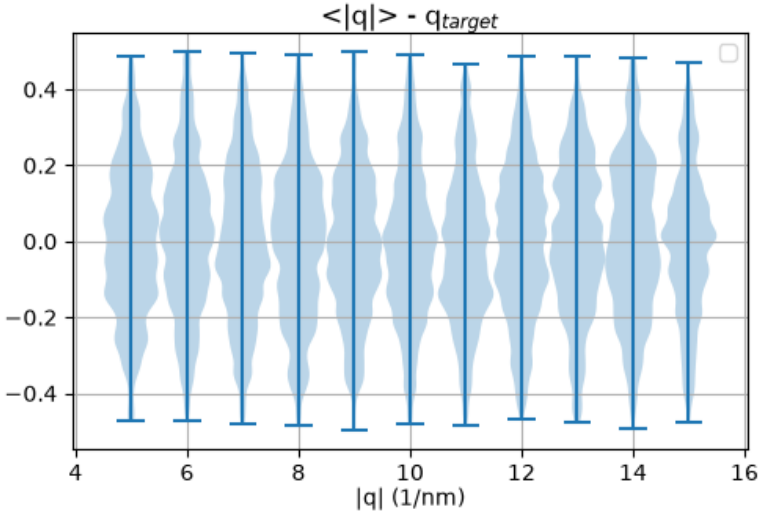
Or, what are all the other vector generators for?

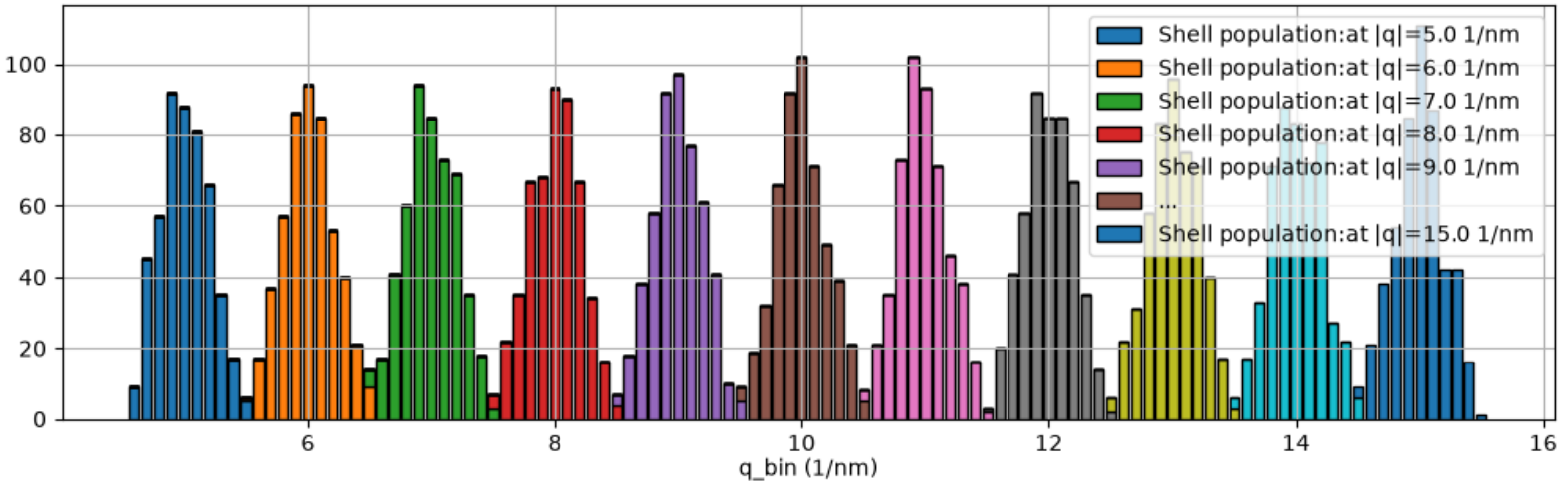


System order		Q vectors symmetry
none (isotropic)	⟷	spherical
1D (nematic)	⟷	circular
2D (layered)	⟷	linear
3D (crystal)	⟷	individual points

How we use Q vectors for isotropic systems

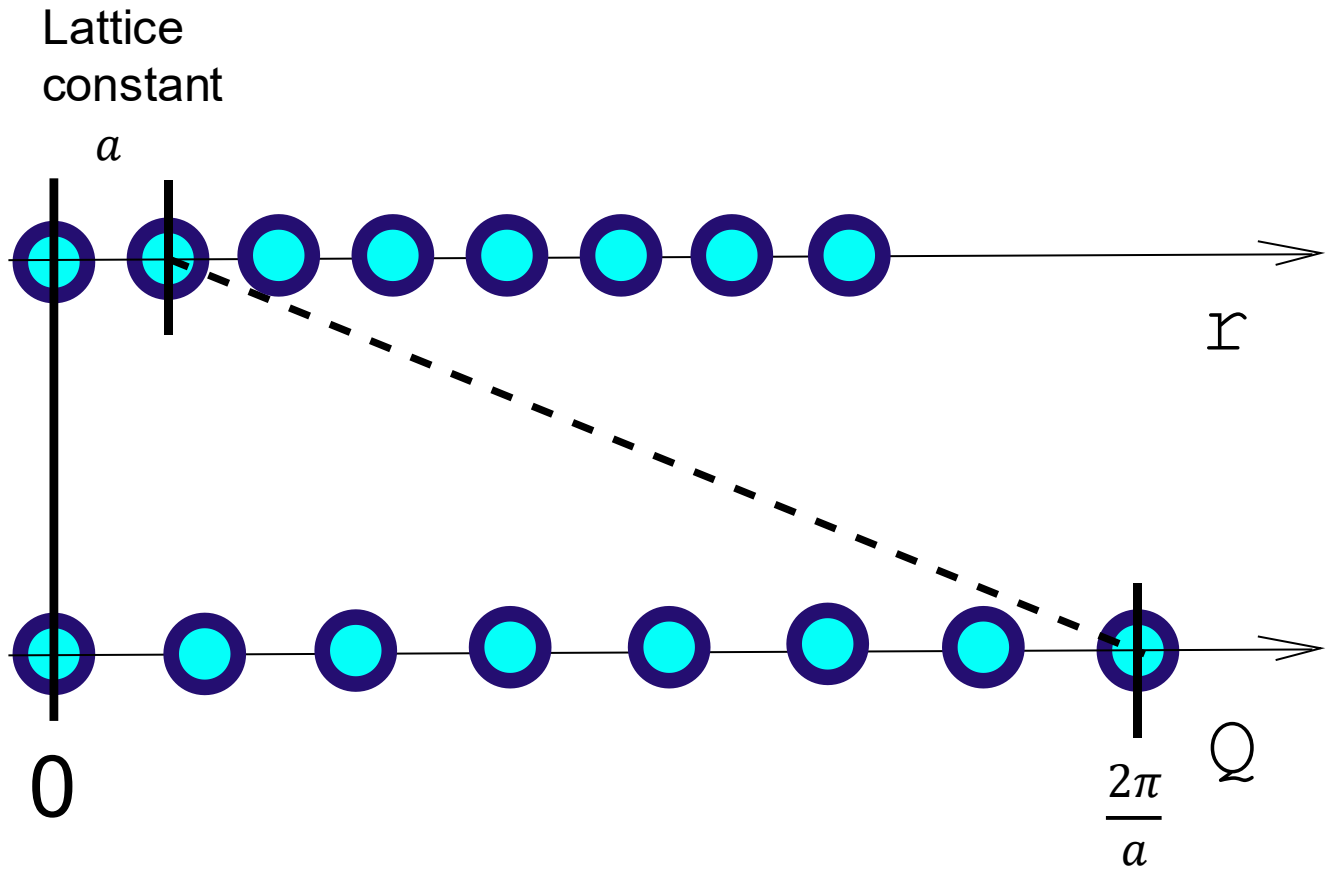






Molecular Dynamics: the boundary conditions

Space limits: size of the simulation box

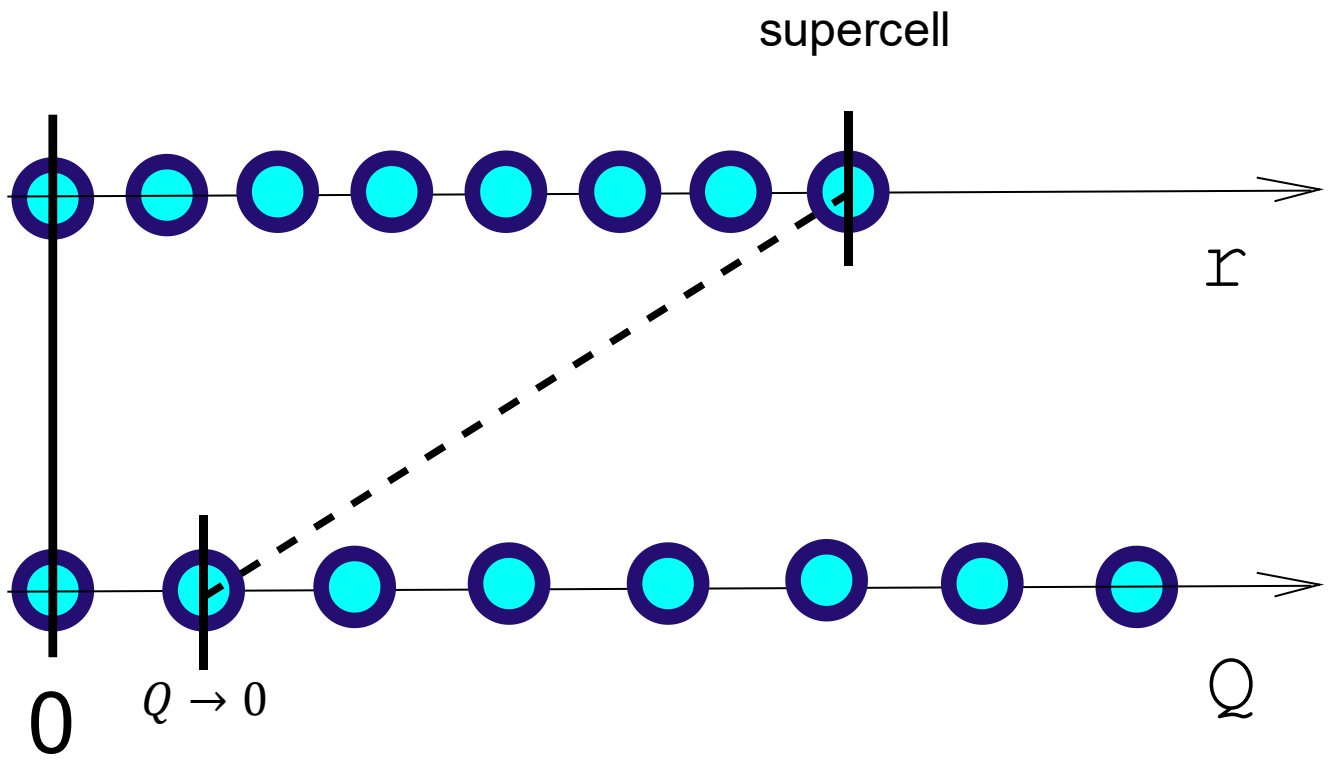


Molecular Dynamics: the boundary conditions

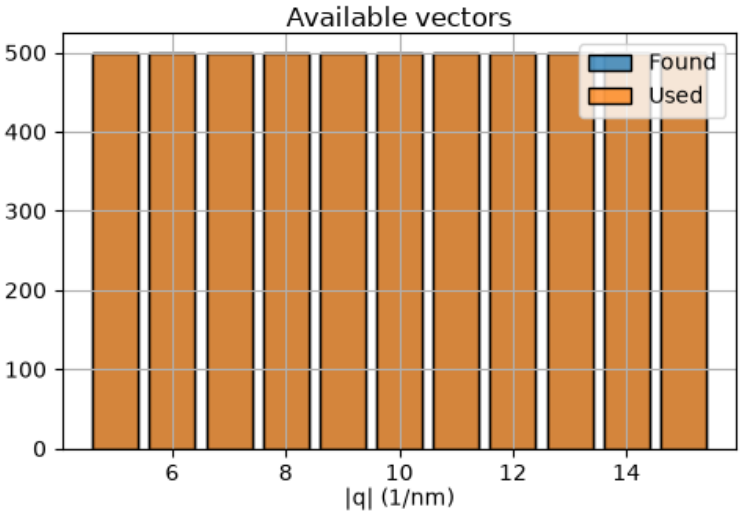
Increasing the size of the system gives access to lower Q values.

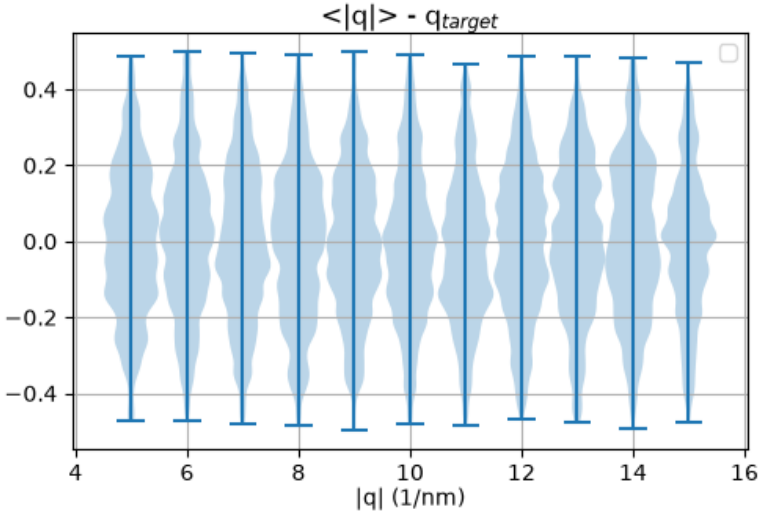
Space limits:

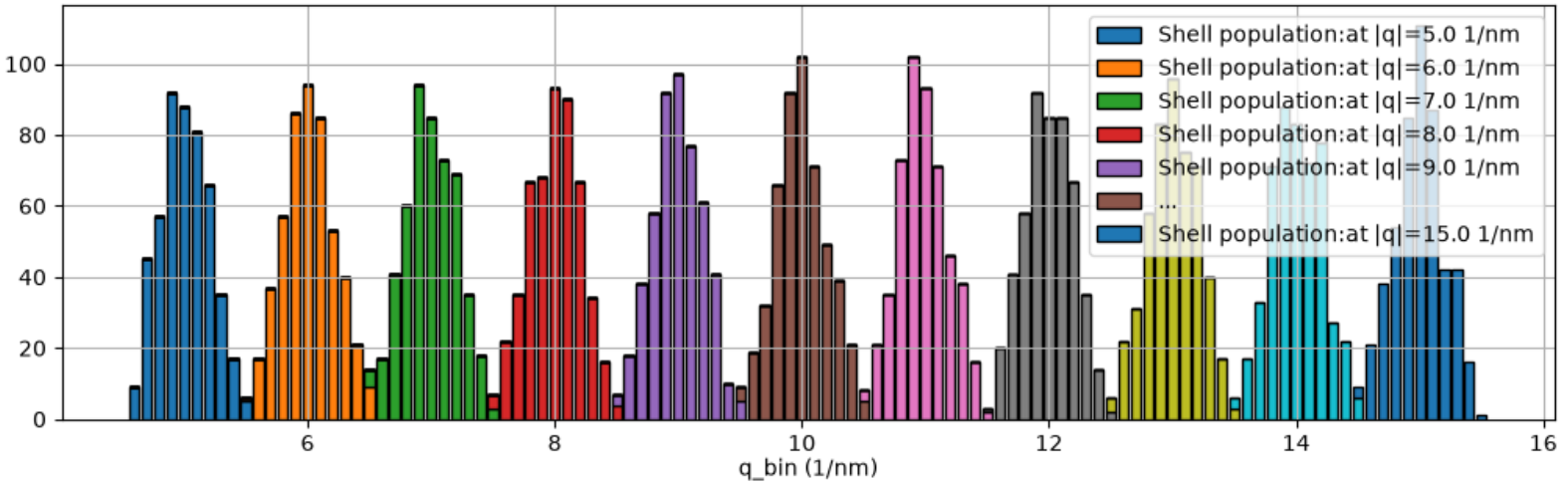
- Size of the simulation box



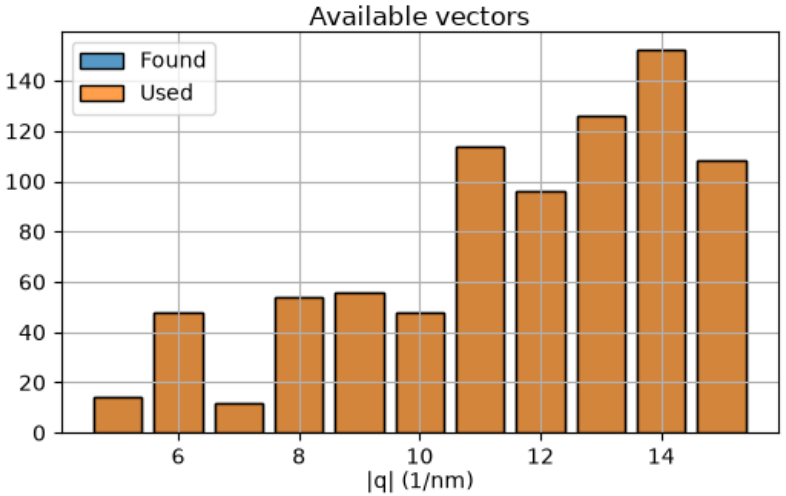
How we use Q vectors for isotropic systems

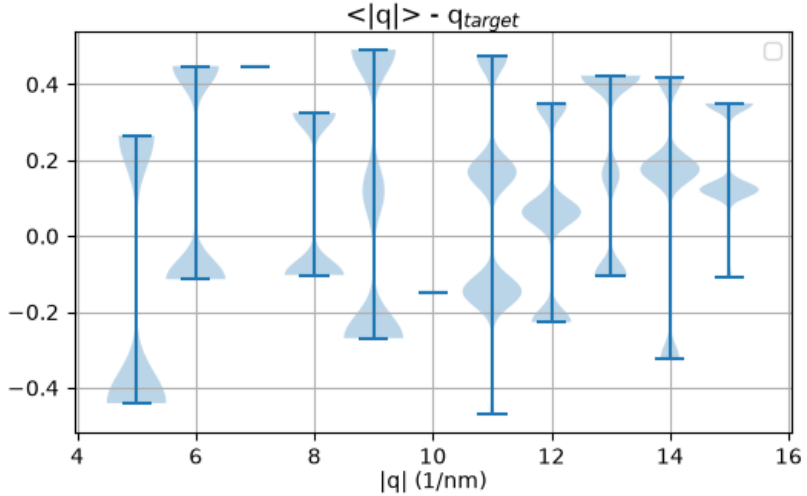


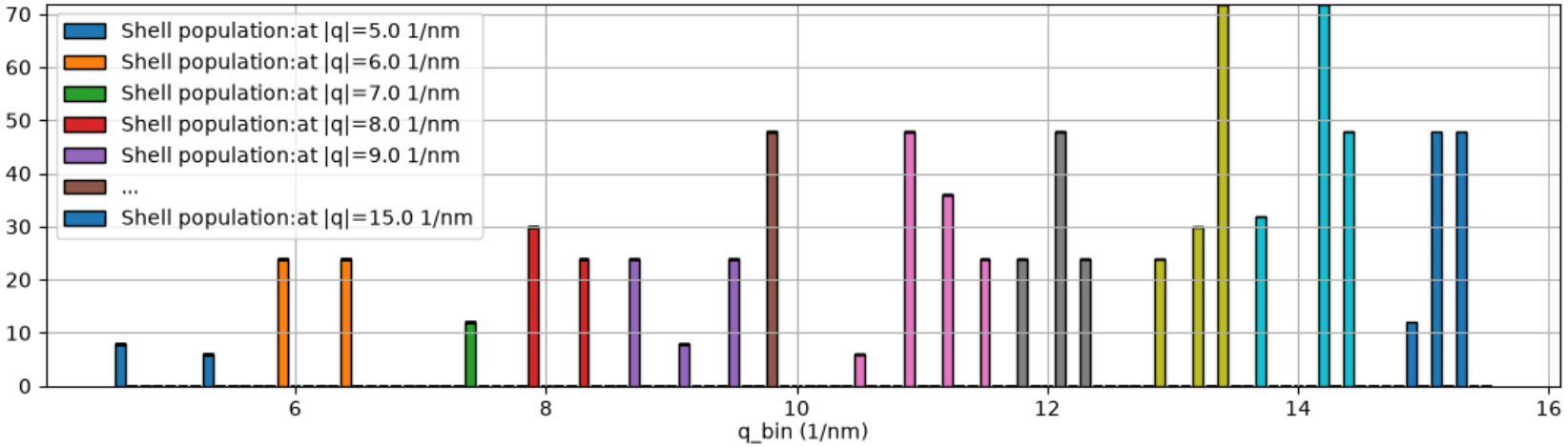




How we use Q vectors for isotropic systems, using only lattice vectors

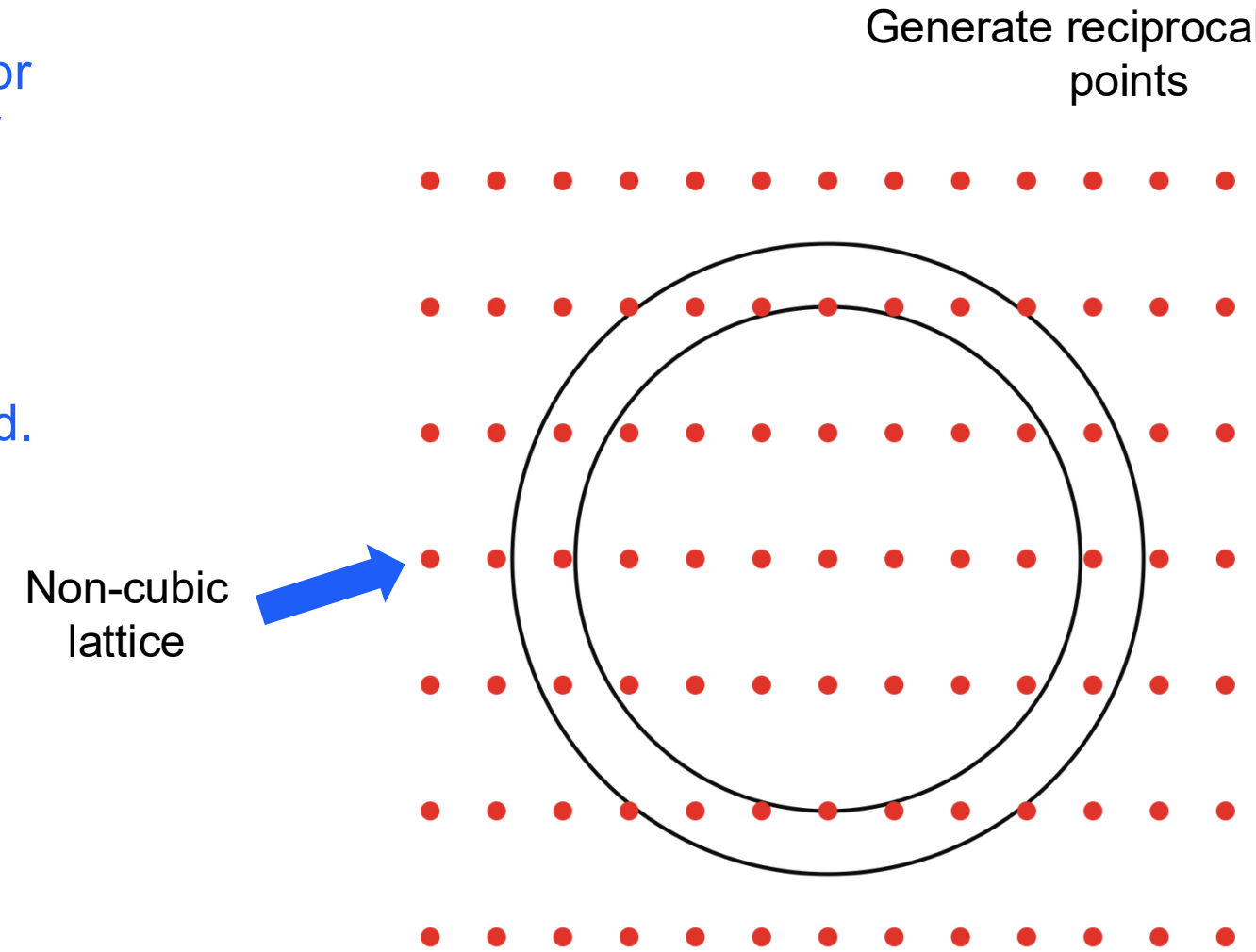






Lattice vectors vs. spherical averaging

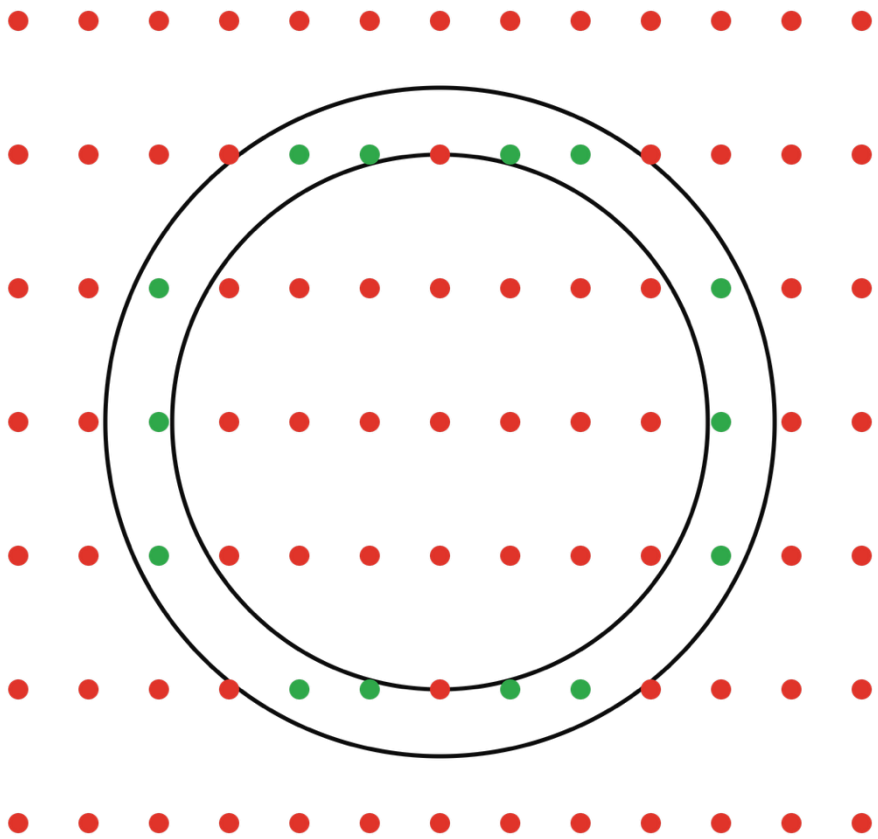
- MDANSE was originally designed for liquid or powdered systems, usually q-vectors will be generated in a spherical shell and results are averaged.
- A new mechanism to improve the spherical averaging has been added.



Lattice vectors vs. spherical averaging

- MDANSE was originally designed for liquid or powdered systems, usually q-vectors will be generated in a spherical shell and results are averaged.
- A new mechanism to improve the spherical averaging has been added.

Select reciprocal lattice points in a shell.

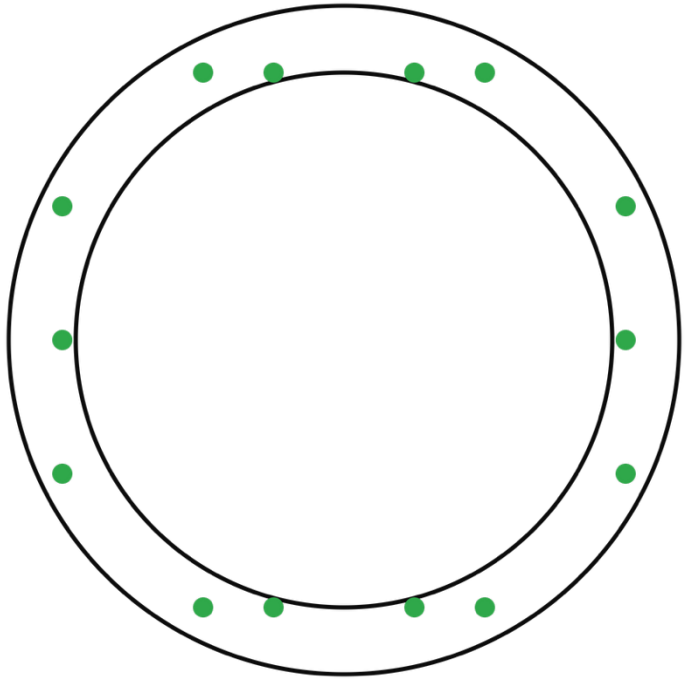


Lattice vectors vs. spherical averaging

- MDANSE was originally designed for liquid or powdered systems, usually q-vectors will be generated in a spherical shell and results are averaged.
- A new mechanism to improve the spherical averaging has been added.
- Old scheme can still be used by setting equal weights.

Old scheme

Calculate S(q, w) averaged over the q-vectors in the shell.

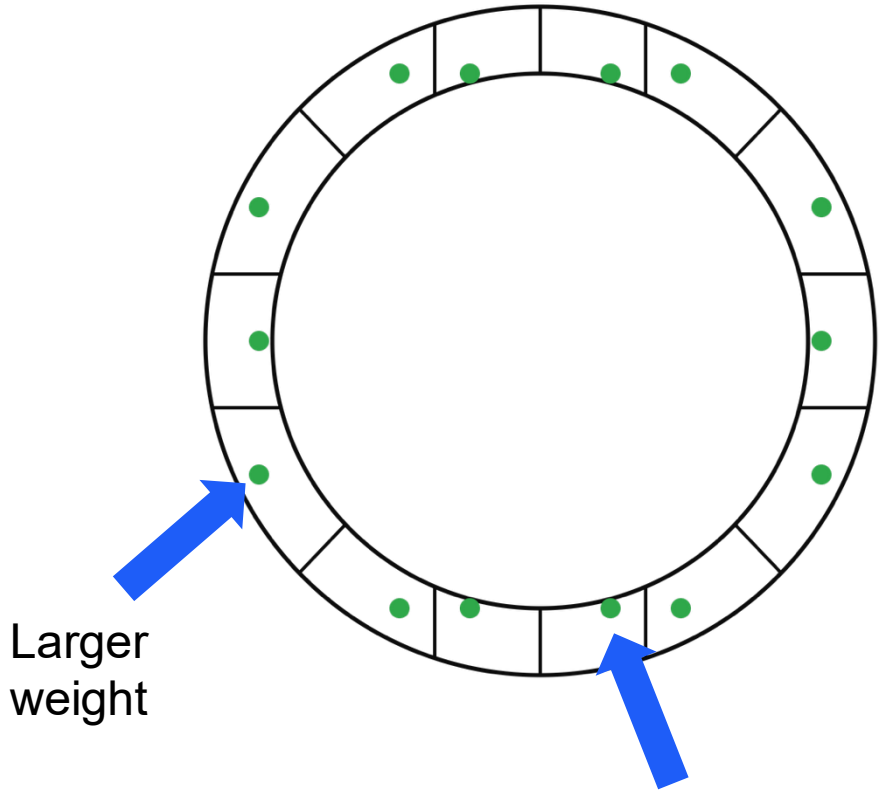


Lattice vectors vs. spherical averaging

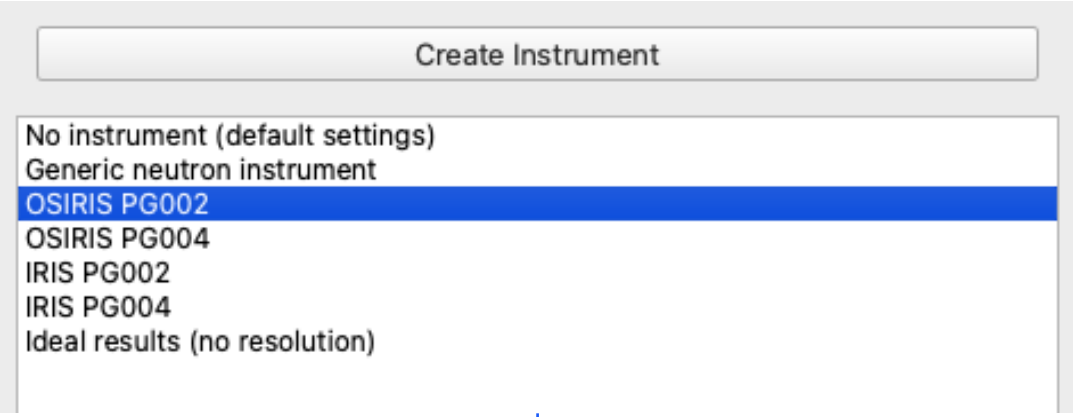
- MDANSE was originally designed for liquid or powdered systems, usually q-vectors will be generated in a spherical shell and results are averaged.
- A new mechanism to improve the spherical averaging has been added.
- Old scheme can still be used by setting equal weights.
- New scheme weights q-vectors based on its shell Voronoi cell volume.

New scheme

Calculate S(q, w) with a weighted averaged over the q-vectors in the shell.



MDANSE implementation of a “neutron instrument”



Basic features of a neutron instrument:

- Resolution (constant, energy independent)
- Coverage (evenly spaced in Q, finite width)

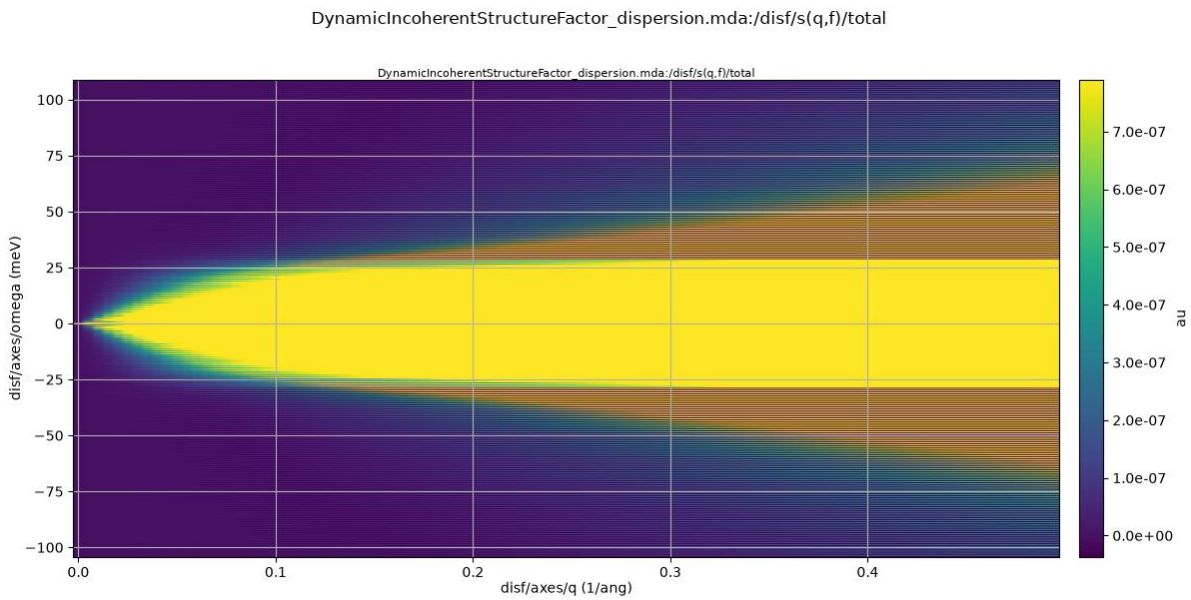
OSIRIS PG002

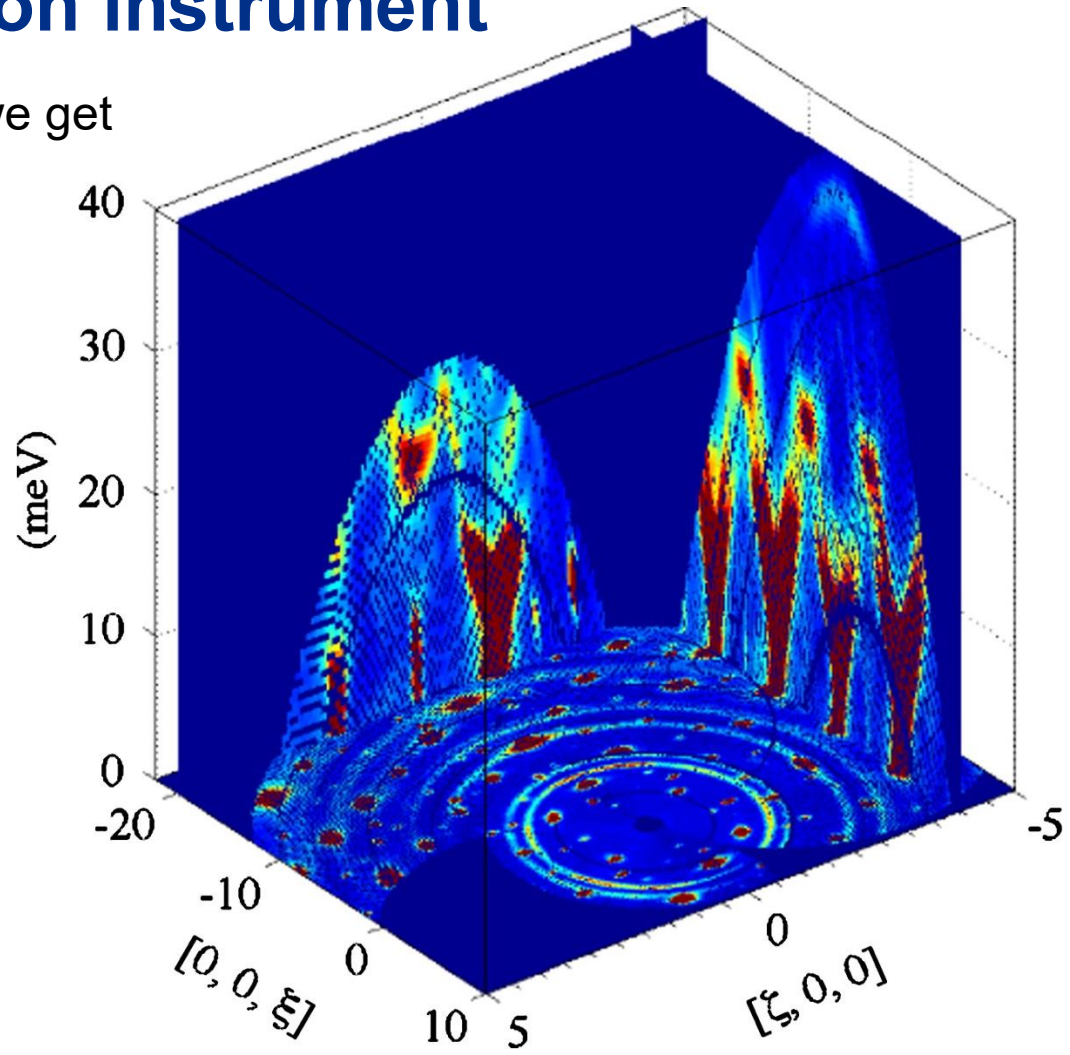
MDANSE resolution input
('gaussian', {'mu': 0.0, 'sigma': 0.01639})
MDANSE q-vector generator input
('SphericalQVectors', {'shells': [1.8, 18.0, 0.5], 'width': 0.2, 'n_vectors': 60})

MDANSE implementation of a “neutron instrument”

In real life, at different Q we get different energy coverage.

In MDANSE, we get full energy range at every Q.





https://doi.org/10.1016/j.nima.2011.01.173
R.I. Bewley et al., Nuclear Instruments and Methods A, Volume 637, Issue 1, (2011) 128-134

Interlude: neutron properties

Wavelength (Ang)	Energy (meV)	Momentum (1/Ang)	Velocity (m/s)	Time of flight 10 m distance (ms)
1	81.787	6.283	3955	2.53
2	20.447	3.142	1978	5.01
3	9.087	2.094	1318	7.6
10	0.82	0.628	396	25.3

Neutron time of flight

Main principle:

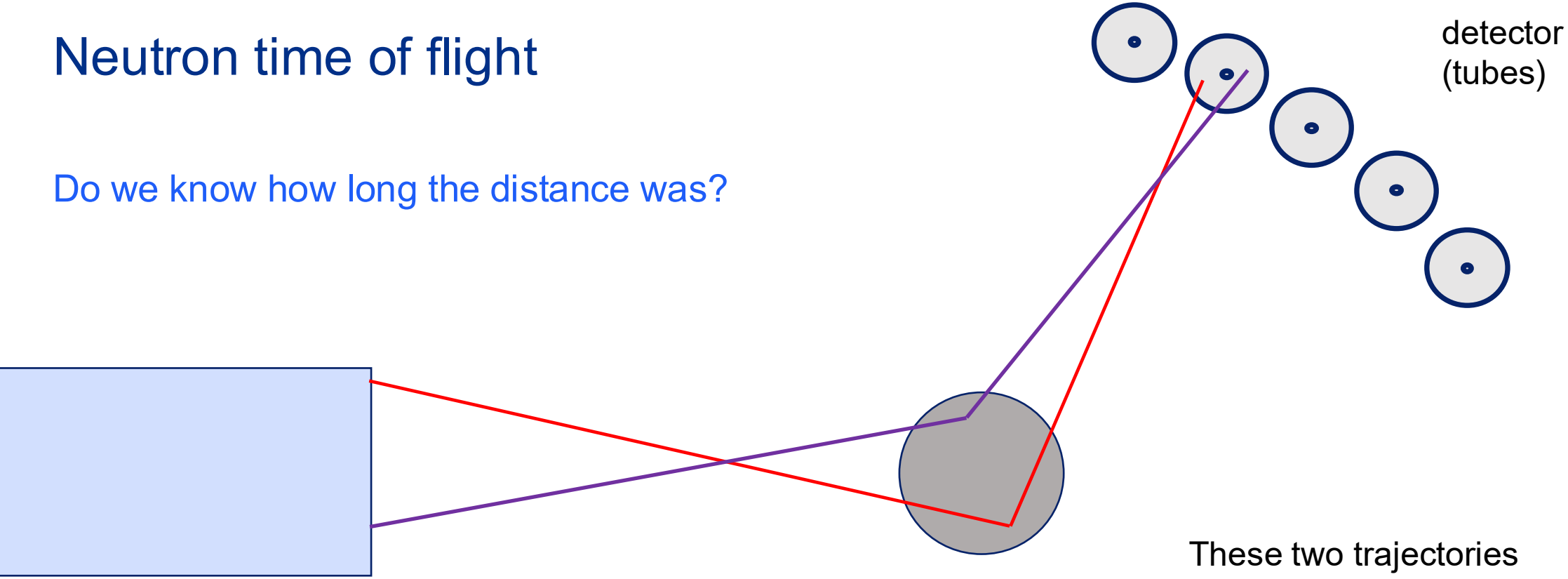
We determine the neutron wavelength (and therefore energy) based on the **time** needed to cover a **distance**.

This is all fine, but:

- do we know how much time the neutron spent between the two points?
- do we know how long the distance was?

Neutron time of flight

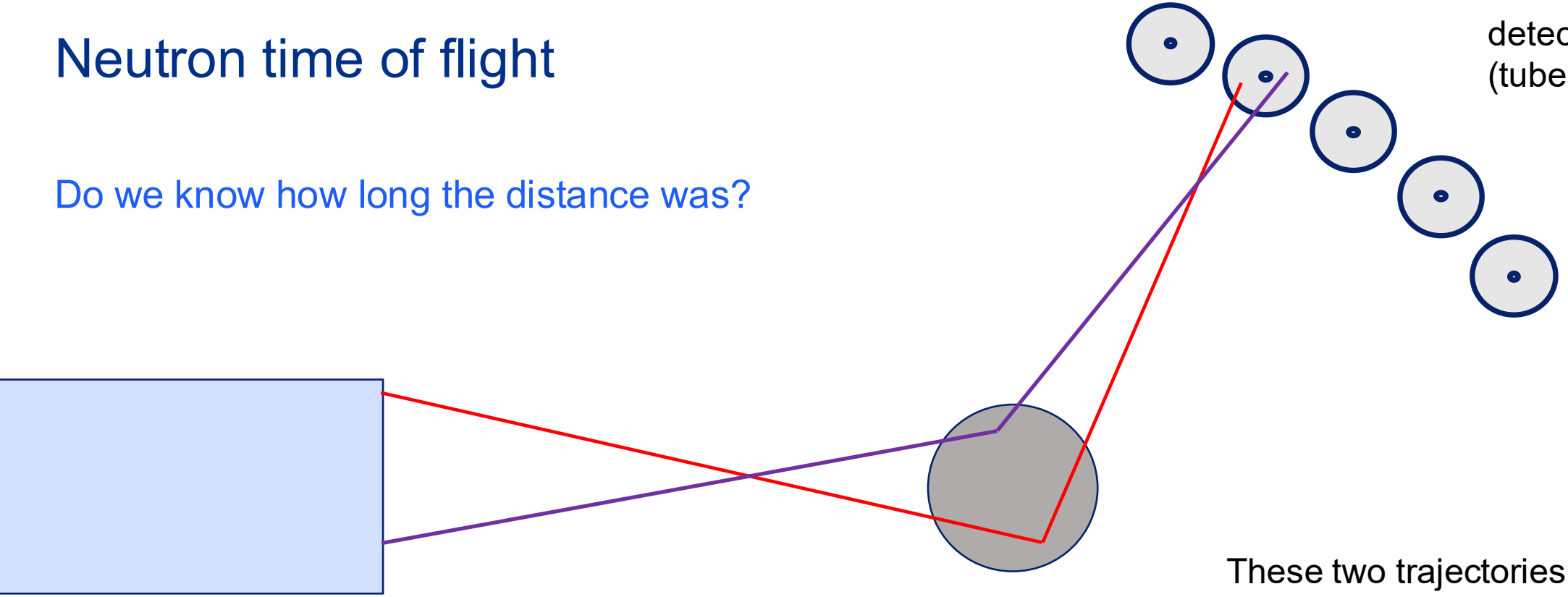
Do we know how long the distance was?



These two trajectories have different **lengths**, but will show up in the same detector pixel.

Neutron time of flight

Do we know how long the distance was?



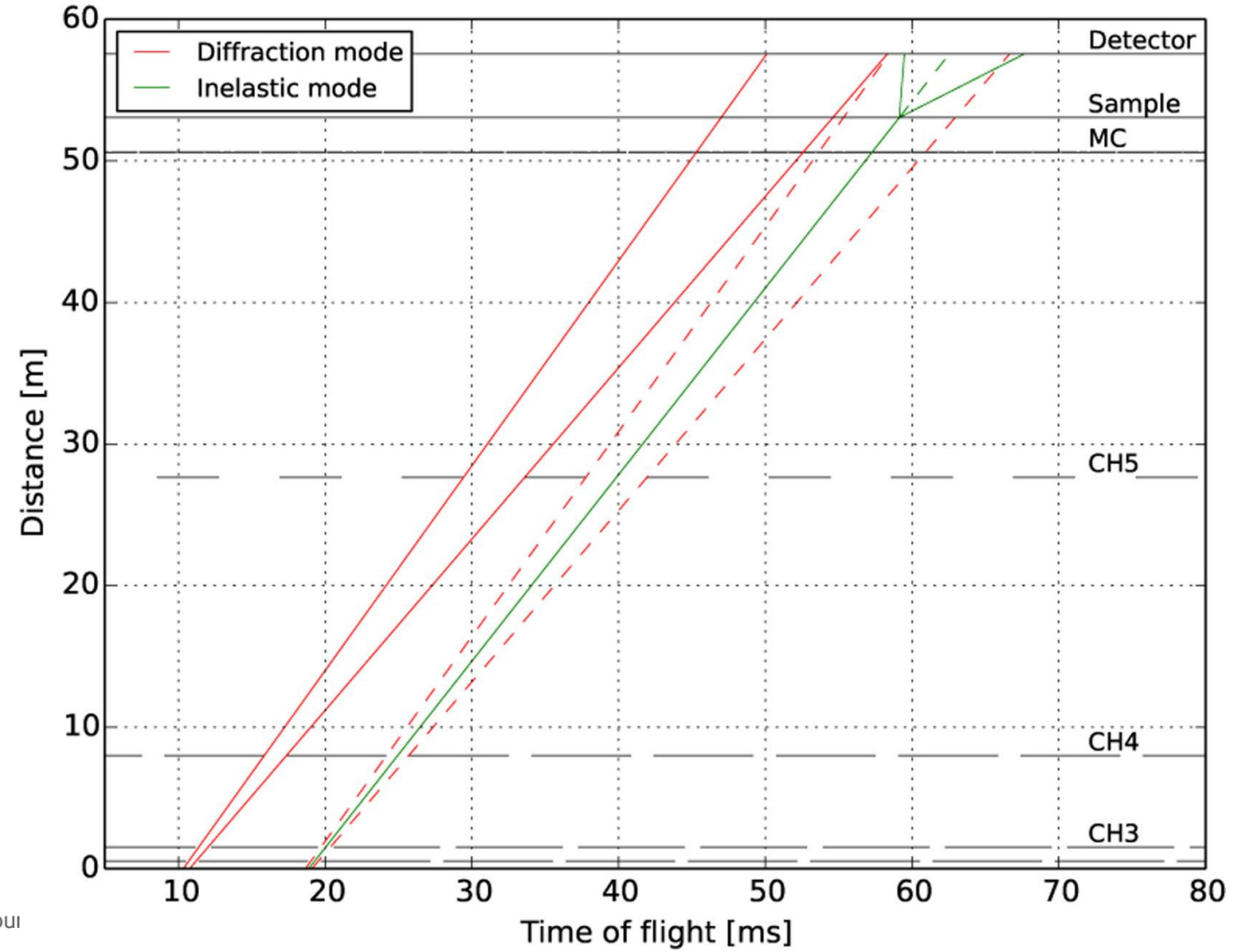
These two trajectories have different **scattering angles**, but will show up in the same detector pixel.

Neutron time of flight

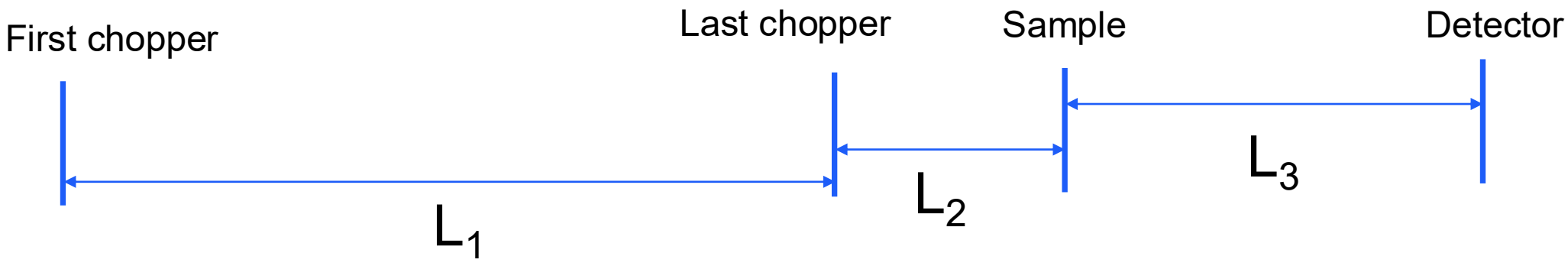
Do we know how much time the neutron spent between the two points?

Faster choppers with smaller windows reduce the time uncertainty.

Turn your TOF diffractometer into a direct TOF spectrometer with this one extra double-disc chopper!



Neutron time of flight



$$\delta E = m_n \left\{ \left(\frac{v_i^3}{L_1} + \frac{v_f^3 L_2}{L_1 L_3} \right)^2 \delta t_p^2 + \left(\frac{v_i^3}{L_1} + \frac{v_f^3 (L_1 + L_2)}{L_1 L_3} \right)^2 \delta t_c^2 + \left(\frac{v_f^3}{L_3} \right)^2 \delta t_d^2 \right\}^{1/2}$$

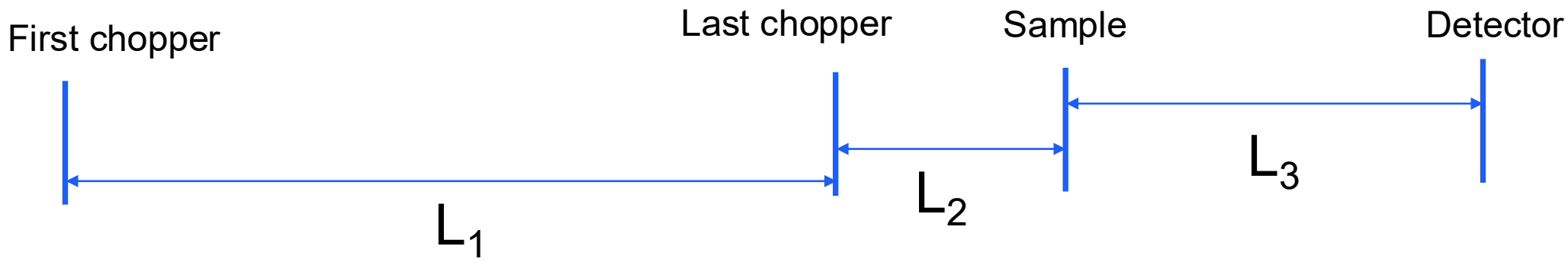
δt_p pulse width of the first chopper

δt_c pulse width of the last chopper

δt_d TOF uncertainty due to sample size and detection volume

CNCS design by G. Ehlers *et al.* https://doi.org/10.1063/1.3626935

Neutron time of flight: direct spectrometer



$$\delta Q_x = \frac{m_n}{\hbar}\left\{\frac{1}{L_1^2}\left(v_i^2 + v_f^2\frac{L_2}{L_3}\cos 2\theta\right)^2 \delta t_p^2 + \frac{1}{L_1^2}\left(v_i^2 + v_f^2\frac{L_1+L_2}{L_3}\cos 2\theta\right)^2 \delta t_c^2 + \left(\frac{v_f^2}{L_3}\cos 2\theta\right)^2 \delta t_d^2 + (v_f \sin 2\theta)^2 (\delta 2\theta)^2\right\}^{1/2}$$

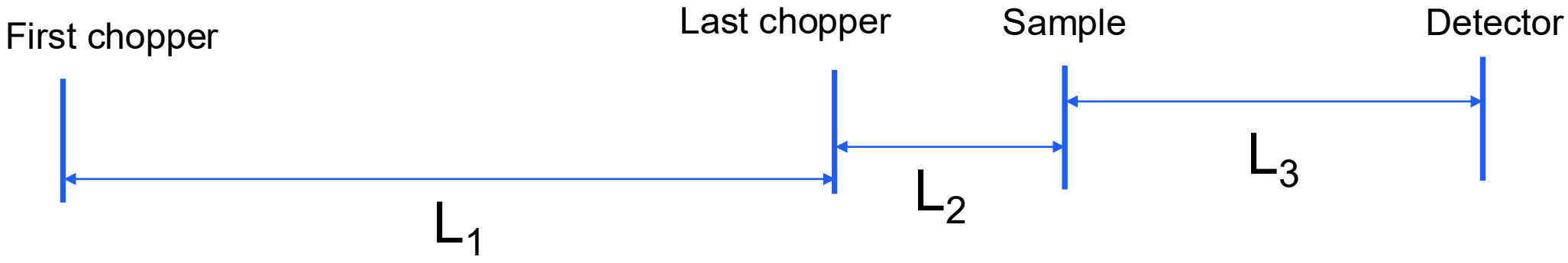
δt_p pulse width of the first chopper

δt_c pulse width of the last chopper

δt_d TOF uncertainty due to sample size and detection volume

CNCS design by G. Ehlers *et al.*
https://doi.org/10.1063/1.3626935

Neutron time of flight: direct spectrometer



$$\delta Q_y = \frac{m_n}{\hbar}\left\{\left(\frac{v_f^2 L_2}{L_1 L_3}\sin 2\theta\right)^2 \delta t_p^2 + \left(\frac{v_f^2}{L_1}\frac{L_1+L_2}{L_3}\sin 2\theta\right)^2 \delta t_c^2 + \left(\frac{v_f^2}{L_3}\sin 2\theta\right)^2 \delta t_d^2 + (v_f \cos 2\theta)^2 (\delta 2\theta)^2\right\}^{1/2}.$$

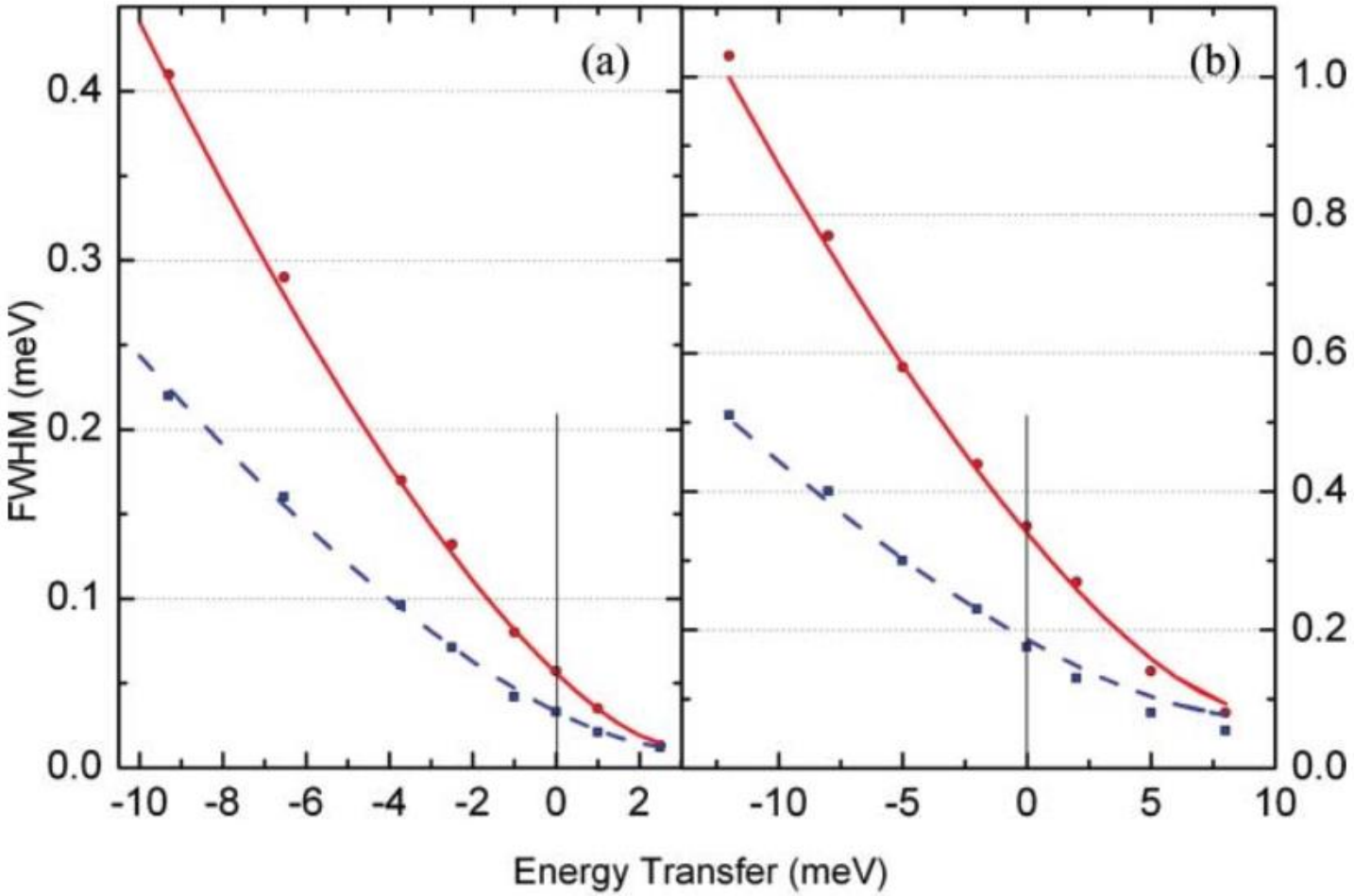
δt_p pulse width of the first chopper

δt_c pulse width of the last chopper

δt_d TOF uncertainty due to sample size and detection volume

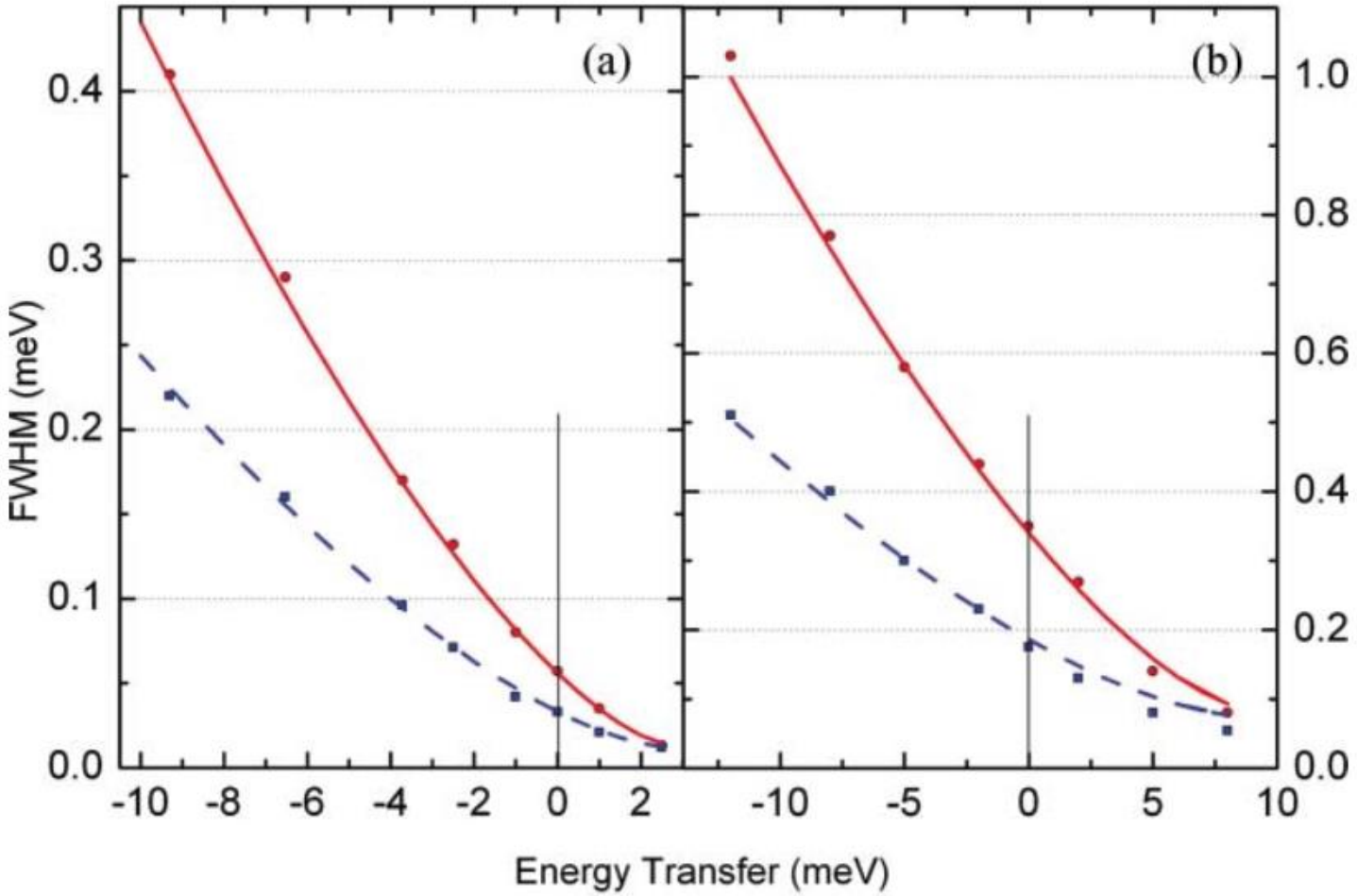
CNCS design by G. Ehlers *et al.*
https://doi.org/10.1063/1.3626935

Neutron time of flight: direct spectrometer



CNCS design by G. Ehlers et al.
https://doi.org/10.1063/1.3626935

Neutron time of flight: direct spectrometer



Using a constant peak width for the entire energy range works only for QENS, where the energy range is small.

MDANSE: ongoing work to start using ResIns for resolution calculations.

CNCS design by G. Ehlers et al.
https://doi.org/10.1063/1.3626935

How can we observe inelastic neutron scattering?

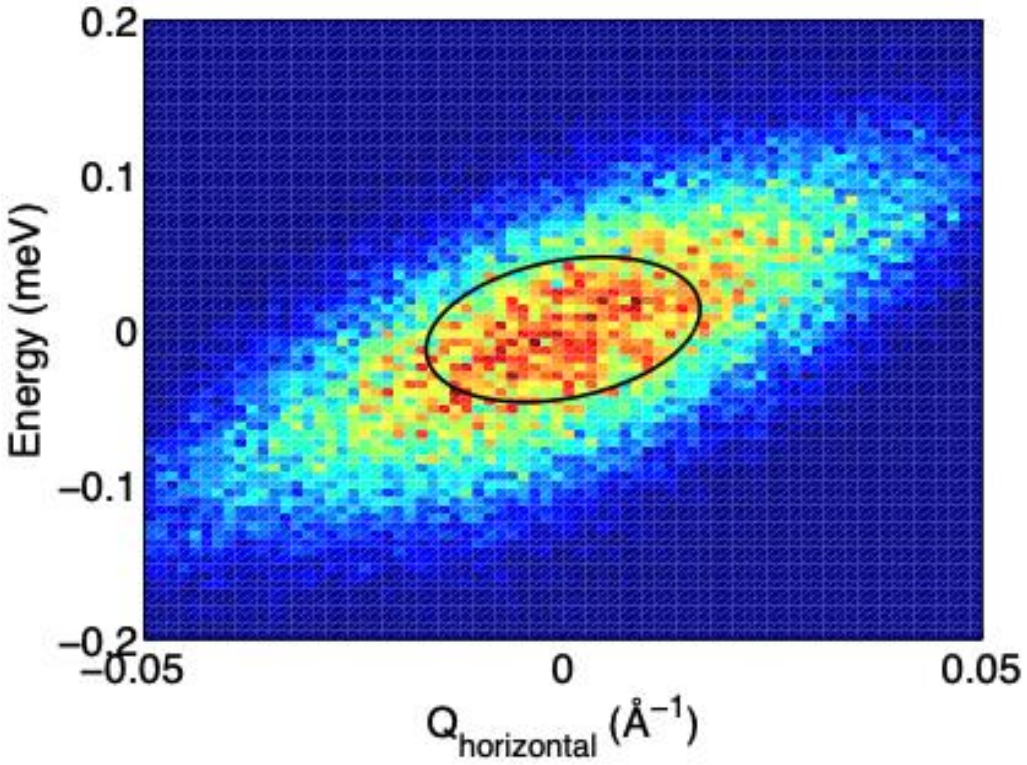
Time of flight instruments:

- direct TOF spectrometer (one wavelength going in)
- indirect TOF spectrometer (one wavelength going out)

Monochromatic instruments:

- triple-axis spectrometers

triple axis spectrometer resolution volume



from https://doi.org/10.1016/j.nima.2013.07.007
(M.D. Le et al., 2013)